Keep, Measure, Erase

What a Quantum Loop Does with Its History

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Quantum loops have a reputation for paradox. We trace the paradoxes to a single fact: the history of a quantum loop, the record of how many times its body ran, is itself quantum information. In a small quantum language in the Π family we can leave this history implicit, keep it, or measure it. The coherent loop is the direct-sum trace of its body: for every finite unitary body it is total and unitary and compiles exactly into loop-free code over a field with decidable equality. A lossless optical network realizes this trace in steady light of one frequency when each return path delays the wave by a whole number of cycles. The other two loops come from the coherent one through two universal constructions on quantum processes: $\mathscr{R}$ adds allocation for a finite counter, and $\mathscr{L}$ adds hiding so we can discard a fresh copy of the guard and feed back the returning outcome. Measuring the guard on every round gives the same data as keeping the exact number of returns and then forgetting it. Forgetting gives the coherent data exactly when the same count state factors out on every input. A finite test on the loop body decides when this holds, so a compiler can safely replace a coherent loop by a measured one even on entangled inputs. Conversely, averaging the coherent loop's output state over a uniformly random return phase yields the measured data, and averaging the circulating weight gives the measured loop's mean number of returns. We implement the language with all three loops and run the paper's examples in it, using a floating-point interpreter and exact checks on real rational tables.

1 Introduction

Can a quantum computer run a loop? The question sounds settled, yet the answers on record seem to contradict one another. Let the loop's guard be in superposition, so that one branch exits after three rounds while another keeps going. If a clock records when each branch halts, branches that leave different records do not interfere [34, 38]. A second approach says the loop must measure its guard on every round, which leaves a classical loop running over quantum data [53]. Such a measurement supplies the classical condition for `while` in Q# and OpenQASM 3 [37, 39]. A third keeps the guard coherent but bounds the rounds and pads shorter branches until all reach the same control state at the same step [55, 56]. A fourth defines the loop as an infinite sum over the paths of its control flow [3, 5]. In our approach, the differences arise from a single design choice, namely what to do with the iteration count:

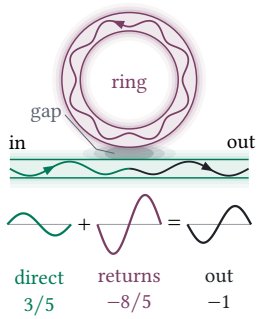
- **Keep it.** Return the count with the data, modulo K for a register of size K.
- **Measure it.** Measure the guard after each call to decide whether to continue.
- **Erase it.** Leave the count implicit so that the exit amplitudes add at the output.

For classical programs the three coincide on the returned data when the count is ignored, which is why classical languages never had to choose. For quantum programs they give three different loop constructs, and a language that provides only one has chosen for its programmers.

Why should the count matter when nobody reads it? A quantum loop can exit after three iterations and after five at once, in superposition, and both exits can reach the same output. There the two contributions combine as complex amplitudes, which behave like signed weights: they can reinforce or cancel, so the final answer depends on both. A counter tags each contribution with its iteration number, and contributions with different tags stay apart whether or not anyone reads the tags. As in Bohr's two-slit example, leaving out the device that records the path can change the answer [8, Fig. 3]. Untagged, the contributions of all iteration counts add into a single map, the infinite sum of the fourth answer. Category theory calls this map the direct-sum trace of the loop body [5, 31], and the geometry of interaction computes it with its execution formula [1]. For a reversible programmer the count is garbage that still carries information: if each round leaves a different tag, discarding the tags gives the same data as measuring the guard on every round.

Leaving the count implicit may look exotic, yet optical devices do it routinely.

A ring resonator is a closed loop of optical waveguide, much like a ring of optical fiber, placed beside a straight waveguide (right). Where the two pass close together, light arriving on the straight guide either continues along it or crosses into the ring, and light already in the ring either goes around again or crosses back out. The waves overlap across the gap, so light can pass between the guides. This junction, called a coupler, plays the role of a loop body with an exit and a way back. Light that leaves after one trip, after two or after a hundred all reaches the same output. Under a steady beam of a single color, these contributions add as amplitudes [2, Eq. (6)] [3, § 5.3], just like the exits of a loop that leaves its count implicit. In this coupler each crossing shifts the phase by a quarter turn, so crossing into the ring and back gives the minus sign in $3/5 - 8/5 = -1$ (Section 4).



We make the choice explicit in ISO, a small quantum language based on reversible pattern matching [12, 42]. A loop body has the shape of the ring's junction. To close it, `trace` feeds the state straight back, count_K also counts the returns modulo K, and `while` measures the guard after each call. The exits on the third and fifth calls have made two and four returns. If we start a bit at zero and flip it on each return, both exits leave it at zero and can still interfere. Measuring the guard on the third call tells them apart, since one exits and the other returns.

- **Coherent or unitary loops compile away.** Solving for the returning state one basis value at a time gives a loop-free program with exact coefficients even when infinitely many iterations contribute [3, proof of Thm. 3.3.15] (Section 3).
- **Keeping and measuring come from two universal constructions.** Allocation supplies the counter and hiding lets the caller discard it [27, 28]. Discarding an exact count gives the same data as `while` (Section 5).
- **Forgetting can be certified.** With m returning basis values and a nonempty input, it suffices to check that the first $m+1$ exit maps are multiples of one map. The measured loop may then replace the coherent one in every channel context (Section 6).
- **The coherent loop can be probed without a count register.** Comparing two nearby phases with one control qubit estimates how much weight circulates in the loop, and averaging that weight over all phases gives the measured loop's mean number of returns (Section 7).

2 Finite programs in ISO

An iso $\omega : a \leftrightarrow b$ is a reversible program between finite types. In Π, 0 has no values, 1 has just $*$, $a + b$ holds a tagged value, and $a \otimes b$ holds a pair [27, § 3.1]. Each basis value $s \in \mathscr{B}(a)$ gives a vector $|s\rangle$. In $1 + 1$ only the tags distinguish the two copies of $*$.

2.1 Syntax

A clause $v \leftrightarrow e$ matches its input against v and evaluates e with the variables it binds. For instance, left x matches left s and binds x to s.

$$
\begin{aligned}
v &::= * \mid x \mid \mathsf{left}\, v \mid \mathsf{right}\, v \mid \langle v_1, v_2 \rangle, \\
p, q &::= * \mid x \mid \langle p, q \rangle, \\
e &::= v \mid \textstyle\sum_i \alpha_i e_i \mid \mathsf{let}\ p = \omega\, q\ \mathsf{in}\ e, \\
\mathscr{T} &::= \{v_i \leftrightarrow e_i\}_{i \in I}, \qquad t ::= e \mid \omega\, t, \\
\omega &::= f \mid \mathscr{T} \mid \omega; \omega \mid \omega \oplus \omega \mid \omega \otimes \omega \mid \omega^\dagger.
\end{aligned}
$$

A clause pattern v can test a tag. In a local call, q assembles the argument and p takes the result apart for e. Because p uses only unit, pairs and variables, it can bind even a tagged value without testing its tag.

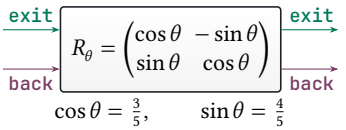
Patterns use distinct variables, and expressions allow local calls and finite sums $\sum_i \alpha_i e_i$ with $\alpha_i \in \mathbb{C}$, written in brackets in the code. We build isos from clause tables and names f of earlier checked definitions. Sequence $f; g$ passes the output of f to g, while $\oplus$ acts on the alternative selected by a tag and $\otimes$ acts on both factors of a pair. Dagger reverses an iso. Section 3 adds feedback.

A closed term t has no free value variables. We write $\omega\, t$ as `omega on t`, so `f; g on t` applies both isos in order. The same spelling applies to isometries and channels in Section 5.

An ordinary definition pairs a signature `f :: a <-> b` with a body `f = omega`. The declaration `round` $f : a + c = \mathcal{T}$ gives f type $(a + c) \leftrightarrow (a + c)$, with the same external type a on both sides. The outer tag is the guard that selects the port: `exit` (left) carries an external value of a, and `back` (right) a return of c. We write `Unit` for 1 and `unit` for $*$. At `Qubit`, or $1 + 1$, `0` and `1` stand for left $*$ and right $*$. Thus `exit 0` expands to left (left $*$), with one tag for the port and one for its payload.

For our first round, let both ports carry `Unit`. Then the two basis inputs are `exit unit` and `back unit`. One call rotates the guard through $\theta = \arccos(3/5)$. In that basis order, the two clauses give the two columns of the matrix beside them.

```
round guardRound : Unit + Unit = {
  exit unit ↔ [ 3/5 . exit unit + 4/5 . back unit];
  back unit ↔ [-4/5 . exit unit + 3/5 . back unit]
}
```



On input `exit unit`, the brackets group the two outputs and the dot scales each by its amplitude. We get $(3/5)|\text{exit}\, *\rangle + (4/5)|\text{back}\, *\rangle$, the first column of R_θ. The `back unit` clause gives the second column in the same order. These clauses give one call. We get a loop in Section 3 by connecting the `back` output to the `back` input.

To evaluate a source iso ω, we unfold its names and replace each checked table by its matrix atom $[M]$. For `guardRound` this gives $[R_\theta]$. The resulting program is a *matrix combinator* built from atoms by the same operations:

$$\kappa ::= [M] \mid \kappa; \kappa \mid \kappa \oplus \kappa \mid \kappa \otimes \kappa \mid \kappa^\dagger.$$

2.2 Statics

To check a table $\omega : a \leftrightarrow b$, we first check that each basis input of a matches exactly one clause and that each output expression has type b under the variables its pattern binds. On basis input s, the clause gives column s of the table's matrix M. Its entry in row t is the amplitude for output t. For `guardRound`, the marked entry $M_{ts} = -4/5$ takes `back` to `exit`.

	exit	s=back
t=exit	3/5	−4/5
back	4/5	3/5

Both columns of `guardRound` have squared norm one, and their inner product is zero: the exit row contributes $-12/25$ and the return row contributes $12/25$ (Figure 1). In general, we check that a and b have equally many basis values and that $M^\dagger M = I$, where dagger takes the conjugate transpose. The diagonal entries of this product are the squared column norms and the off-diagonal entries are the inner products of distinct columns. A square matrix that passes this check is unitary.

The judgment $\Sigma; \Delta \vdash e : b$ gives e output type b in the earlier iso definitions Σ and the context of value variables Δ. In a pair pattern $\langle v_1, v_2 \rangle$, linear use splits Δ between the components. A superposition offers alternatives, so each alternative uses the same variables once. A local call splits variables between argument and continuation, where its result pattern binds fresh variables.

The judgment $\Sigma \vdash \omega : a \leftrightarrow b$ checks tables and matrix atoms by the same column test, which is decidable over fields such as $\mathbb{Q}(i, \sqrt{2})$. For a composite we check that the interfaces match. Checking a table evaluates its clauses, but each local call uses an earlier definition or a smaller subprogram, so both checking and evaluation terminate (Appendix A) [42, §§ 2, 5] [33, Ch. 3].

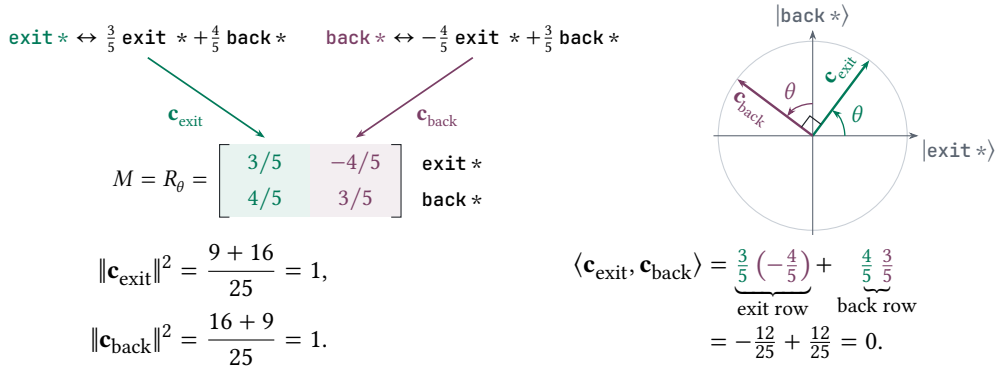


Fig. 1. Each clause gives one column of R_θ. The rotation turns each basis vector through the same angle θ, so the columns lie on the unit circle and remain perpendicular. Each product in their inner product uses entries from the same output row. For complex columns we conjugate the first entry of each product.

2.3 Dynamics

To compute a column in Figure 1, we match a basis input against its clause. With the resulting bindings γ, a value v gives $|v[\gamma]\rangle$ and a sum gives the weighted sum of its vectors. A local `let` uses each basis output of the called iso to bind the variables in the continuation, then adds the resulting vectors with their amplitudes. We write $\gamma \vdash e \Downarrow \psi$ for the result.

For `guardRound`, elaboration gives the atom $[R_\theta]$ and evaluation returns its matrix R_θ. In general, if elaboration gives $\mathscr{E}_\Sigma(\omega) = \kappa$ and $\kappa \Downarrow M$, we write $\Sigma \vdash \omega \Downarrow M$. Applying the program to a closed input $t \Downarrow \psi$ then gives $\omega\, t \Downarrow M\psi$.

When we apply `guardRound` twice to `exit unit`, the first call places $3/5$ on `exit` and $4/5$ on `back`. The second call therefore uses both clauses, weighting their columns by those amplitudes:

$$\frac{3}{5}\begin{pmatrix}3/5\\4/5\end{pmatrix} + \frac{4}{5}\begin{pmatrix}-4/5\\3/5\end{pmatrix} = \begin{pmatrix}-7/25\\24/25\end{pmatrix}.$$

In sequence the whole first output enters the second call. Thus if $\kappa \Downarrow M$ and $\lambda \Downarrow N$, then $\kappa;\lambda \Downarrow NM$, with M acting first. Elaboration also preserves sum, tensor and dagger, which take the direct sum, tensor product and conjugate transpose of the matrices.

2.4 Denotation

A type a denotes $[\![a]\!] = \mathbb{C}^{\mathscr{B}(a)}$, with one orthonormal vector per basis value. Thus 0 and 1 denote the zero space and $\mathbb{C}$, while sums and products denote direct sums and tensor products.

A checked iso $\omega : a \leftrightarrow b$ denotes the linear map $[\![\omega]\!] : [\![a]\!] \to [\![b]\!]$ that agrees with its clauses on basis inputs. For `guardRound`, amplitudes u on `exit` and v on `back` therefore weight the two columns of the rotation.

$$[\![\texttt{guardRound}]\!]\begin{pmatrix}u\\v\end{pmatrix} = u\begin{pmatrix}3/5\\4/5\end{pmatrix} + v\begin{pmatrix}-4/5\\3/5\end{pmatrix} = \begin{pmatrix}(3u-4v)/5\\(4u+3v)/5\end{pmatrix}.$$

The column check makes this map unitary (Appendix A gives the correspondence with evaluation). In sequence we pass both output coordinates to the next call. To form a loop we let the `exit` amplitude leave and feed back only the `back` amplitude.

3 What the coherent loop computes

Closing `guardRound` feeds the lower output back into the next call. The upper output carries the first $3/5$ to the caller, where it adds to the exits from later calls.

3.1 From a round to its exits

Follow the $4/5$ on the lower wire in Figure 2. The second call sends $(4/5)(-4/5) = -16/25$ to the upper port and $(4/5)(3/5) = 12/25$ back for a third call. A classical reversible body follows one path on each basis input. Here several exits contribute. We number the trips to work out what each contributes. Those numbers let us calculate the sum without putting a counter in the program. Only the sum reaches the caller.

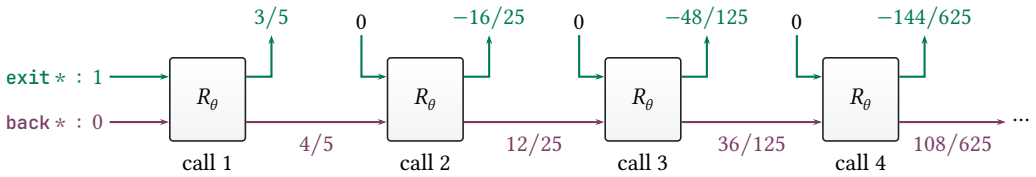


Fig. 2. Four calls to `guardRound`. The upper outputs contribute to the caller's result; the first two add to $-1/25$. The lower wire carries the returning amplitude to the next call, whose external input is zero. Over all calls the upper outputs sum to $3/5 - 16/25 - 48/125 - 144/625 - \cdots = -1$.

Syntax and statics. Elaborating `trace guardRound` replaces the clauses by $[R_\theta]$ and gives the matrix combinator $\mathrm{trace}_1[R_\theta]$. The two typing rules close the same wire before and after elaboration:

$$\underbrace{\frac{\Sigma \vdash \omega : a + c \leftrightarrow b + c}{\Sigma \vdash \mathrm{trace}_c\, \omega : a \leftrightarrow b}}_{\text{source iso}} \qquad \underbrace{\frac{\kappa : a + c \leftrightarrow b + c}{\mathrm{trace}_c\, \kappa : a \leftrightarrow b}}_{\text{matrix combinator}}.$$

Elaboration leaves the return wire in place as it translates the body, so $\mathscr{E}_\Sigma(\mathrm{trace}_c\, \omega) = \mathrm{trace}_c\, \mathscr{E}_\Sigma(\omega)$. In `trace f; g` we close `f` first, then apply `g`. Unitarity allows distinct external types a, b of equal dimension. A `round` uses the same type on both sides, giving `trace guardRound` type $1 \leftrightarrow 1$. Closing just the return port gives the interface of qWhile's do-while combinator [3, § 5.1] and implements Lemonnier's proposal to use Bartha's trace for reversible programs on finite data [33, § 5.2.3].

Each box in Figure 3 is a morphism in Unitary, the category of finite-dimensional Hilbert spaces and unitary maps. For a common payload P, $F_g : P \oplus P \to \mathbb{C}^2 \otimes P$ puts guard and payload on one sheet. Its inverse separates the two alternatives again so we can feed back just the return.

The exit maps. For a body $\beta : a + c \leftrightarrow b + c$, write $X = [\![a]\!]$, $X' = [\![b]\!]$ and $Y = [\![c]\!]$. Group the columns of its matrix U by input tag and the rows by output tag. On external input u and return y,

$$U\begin{pmatrix} u \\ y \end{pmatrix} = \begin{pmatrix} A & B \\ C & D \end{pmatrix}\begin{pmatrix} u \\ y \end{pmatrix} = \begin{pmatrix} Au + By \\ Cu + Dy \end{pmatrix}. \tag{3.1}$$

The top row gives the output in X' and the bottom row the return in Y. In `guardRound`, the first clause gives $A = 3/5$, $C = 4/5$ and the second gives $B = -4/5$, $D = 3/5$. The blocks $A : X \to X'$, $B : Y \to X'$, $C : X \to Y$ and $D : Y \to Y$ are morphisms in **FdHilb**, the category of finite-dimensional Hilbert spaces and linear maps. Write $m = \dim Y$ for the number of returning basis values. Since U is unitary and Y occurs on both sides, the external basis sizes also agree: $n = \dim X = \dim X'$.

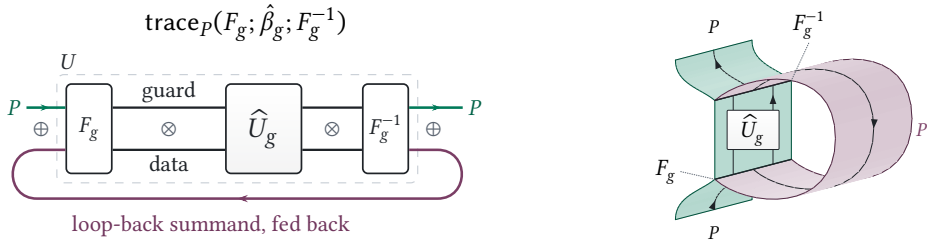


Fig. 3. Read the circuit rightward and the sheets upward. Separate sheets carry alternatives (⊕), and wires on one sheet carry pairs (⊗) [15, Defs. 16, 18–20, 22]. The middle box is the matrix $\widehat{U}_g$ of $\hat{\beta}_g = F_g^{-1}; \beta; F_g$, with ; read left to right. Splitting off the return at F_g^{-1} and feeding it back gives the direct-sum trace $P \to P$ [5, 31].

Nothing has returned on the first call, so $(u, 0)$ gives the immediate exit Au and puts Cu on the return wire. The next call takes $(0, Cu)$ to (BCu, DCu). Write $x_k : X \to X'$ for the exit map after k returns. To reach x_{k+1}, amplitude enters through C, passes through D another k times, then leaves through B. Thus

$$x_0 = A, \qquad x_{k+1} = BD^kC. \tag{3.2}$$

After $k+1$ calls, $x_0u, \dots, x_ku$ have reached the external output and D^kCu remains on the wire. Adding all the exits gives the execution formula $[\![\mathrm{trace}_c\, \beta]\!] = L = \sum_{k\geq 0} x_k$ [1, 19]. In the ring, x_0 gives the direct path and x_k the light that made k trips.

To sum these exits we need to know how much amplitude from u can still be circulating after many calls. If $Dv = \lambda v$ with $|\lambda| = 1$, the whole norm of v returns, so $Bv = 0$. These states span a space Y_0 on which D is unitary. An external input is orthogonal to Y_0 and stays so under U. Its returns therefore lie in $Y_1 = Y_0^{\perp}$, where the powers of D decay geometrically.

3.2 Computing the sum without a last round

On input `unit`, `guardRound` sends $4/5$ back on its first call and $3/5$ of what it receives on each later call (Figure 2). If y is the sum of the returns, the terms after the first sum to $(3/5)y$.

$$y = \frac{4}{5} + \underbrace{\frac{12}{25} + \frac{36}{125} + \frac{108}{625} + \frac{324}{3125} + \cdots}_{(3/5)y} = \frac{4}{5} + \frac{3}{5}y.$$

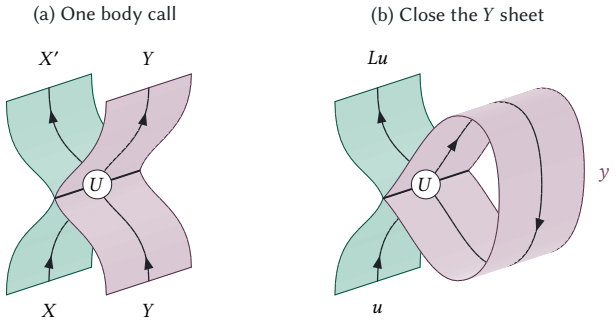
This is the *wire equation* for `guardRound`. In general, Cu first enters the return wire and each later call applies D, so the sum satisfies $y = Cu + DCu + D^2Cu + \cdots = Cu + Dy$. The caller receives the immediate exit Au together with By, as Figure 4 illustrates.

Theorem 3.1 (Finite feedback). *For a finite unitary body, Y_0 and $Y_1 = Y_0^{\perp}$ reduce D, C maps into Y_1, and the powers of $D|_{Y_1}$ decay geometrically. The exits sum absolutely to a unitary L and satisfy $\sum_k x_k^{\dagger} x_k = I_X$. Every input u admits a solution of $y = Cu + Dy$, and each solution gives $Lu = Au + By$.*

The first call splits $\|u\|^2$ into $\|Au\|^2 + \|Cu\|^2$. The next splits $\|Cu\|^2$ into $\|BCu\|^2 + \|DCu\|^2$. Since the remaining weight tends to zero, the squared exit norms sum to $\|u\|^2$. Closing the wire gives $U(u, y) = (Lu, y)$, so canceling $\|y\|^2$ shows $\|Lu\| = \|u\|$. Equal external dimensions make L unitary, as in Bartha's finite trace theorem [5, Thms. 4, 7–8]. Appendix B proves the decay bound.

For each basis input $|s\rangle$, put the sum of the returns in column s of F. The wire equations then read $(I_Y - D)F = C$, and column s of $A + BF$ gives the output on $|s\rangle$.

$$\frac{\kappa \Downarrow U \quad U = \left(\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right) \quad F : X \to Y \quad (I_Y - D)F = C}{\mathrm{trace}_c\, \kappa \Downarrow A + BF}.$$



Both ends of the purple sheet carry the same amplitude y. For $u = 1$ the wire equation gives

$$\tfrac{2}{5}y = \tfrac{4}{5}, \qquad y = 2.$$

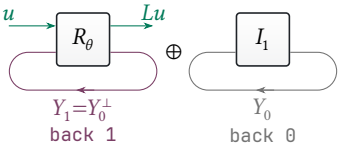
The green output receives $3/5$ directly and $-4/5$ of each return, giving $3/5 - (4/5)y = -1$. Thus

$$R_\theta \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} -1 \\ 2 \end{pmatrix}.$$

Fig. 4. In Unitary, the open body at left is $U : X \oplus Y \to X' \oplus Y$. Joining the two ends of the purple Y sheet at right gives its direct-sum trace $L = \mathrm{Tr}^Y(U) : X \to X'$.

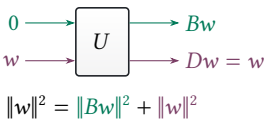
An undetermined return. For `guardRound` we divided by $1 - 3/5$. To see why $I_Y - D$ can have no inverse, keep the original return on `back 1` and let a second value, `back 0`, return to itself.

```
round extraPort : Unit + Qubit = {
  exit unit ↔ [ 3/5 . exit unit + 4/5 . back 1];
  back 1    ↔ [-4/5 . exit unit + 3/5 . back 1];
  back 0    ↔ back 0
}
```



Here `back 0` spans the trapped space Y_0 of Theorem 3.1, and `back 1` spans its orthogonal complement $Y_1 = Y_0^\perp$. The rotation gives $y_1 = \frac{4}{5}u + \frac{3}{5}y_1$, so $y_1 = 2u$. The identity gives $y_0 = y_0$ for every y_0, so $I_Y - D = \mathrm{diag}(2/5, 0)$ has no inverse. Starting from u gives $y_0 = 0$ because no call reaches `back 0`. Only `back 1` feeds the exit, so every choice of y_0 gives $(3/5)u - (4/5)y_1 = -u$.

For input u, the sum of the returns gives one solution y. If $y + w$ is another, the outputs differ by Bw. Subtracting the wire equations cancels the common first return Cu, leaving $Dw = w$. As the inset shows, one call returns the whole difference unchanged. Since U preserves norm, nothing remains at the exit ($Bw = 0$), so both solutions give the caller $Au + By$.



So far our wire has fed amplitude from one call to the next. Czachor uses the same connection to model a wormhole, feeding one port's later output back to its earlier input [16, Eqs. (6)–(7)]. The grandfather paradox imagines a traveler who goes back and prevents their own departure. At the entrance to Czachor's wormhole, the incoming wave meets the wave that has gone around the loop. For a lossless loop with $I_Y - D$ invertible, their interference gives a unitary external map. To a traveler who tries to enter, it is as if the universe had placed a mirror across the doorway: the traveler is reflected with probability one. A perfect absorber inside the wormhole cuts off the wave that made the entrance act like a mirror. If the entrance sends some of the incoming wave into the loop, the traveler now has a chance to step through. If they do, it is a one-way ticket: the absorber stops them before they can complete the trip [16, Thm. 1 and § IV].

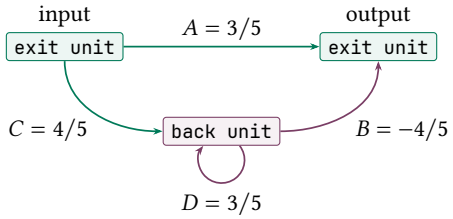
Hawking and Visser ask what could prevent a time machine from forming [25, 47]. In Czachor's model the wormhole already exists. He calls reflection a form of *chronology protection*: for such a loop, unitarity alone forces reflection [16, § I].

When a single complex amplitude returns unchanged, Czachor observes that its port is isolated, just like `back 0`. He asks what happens with more return ports [16, arXiv v8, after Eq. (45)]. Bartha's

work on quantum Turing automata gives the finite answer: he proved that the paths to the exit sum to an isometry for any finite isometric body, even if $I_Y - D$ is singular [5, Eq. (1), Thm. 8]. Here $Bw = 0$ shows why the caller gets the same output from any solution.

3.3 Compiling away the loop

Building on Bartha's proof, Andrés-Martínez eliminates one return value at a time for finite contractions [3, Thm. 3.3.15, Cor. 3.3.16, pp. 165–166]. We use that construction to write the clauses of $L = A + BF$. In `guardRound` the first clause sends $3/5$ to `exit unit` and $4/5$ to `back unit`. To replace `back unit` in that clause, we sum the exits from that port.



On input `back unit`, each further return contributes a factor D. The exits sum to

$$B + BD + BD^2 + \cdots = \frac{B}{1 - D} = -2.$$

The first clause sends $C = 4/5$ there, giving $(4/5)(-2)$. The wire equation groups the same factors as $BF = (-4/5) \cdot 2$, with $F = C/(1 - D)$.

In the first clause we replace `back unit` by $-2\,(\texttt{exit unit})$. Both terms now leave by the same port, so we can add their coefficients and drop the external tag.

$$\texttt{exit unit} \;\leftrightarrow\; \tfrac{3}{5}\,(\texttt{exit unit}) + \tfrac{4}{5}\big(-2\,(\texttt{exit unit})\big) = -\texttt{exit unit} \qquad \rightsquigarrow \qquad \{* \leftrightarrow -1 \cdot *\}.$$

In `flipRound` (Appendix F.1), each call rotates the guard and flips a data bit, so neither `back 0` nor `back 1` returns to itself in one call. Follow `back 1` in Figure 5. It can return to `back 0` and then to `back 1`, with amplitude $3/5$ each way. Removing `back 0` turns those two steps into a single return with amplitude $9/25$. We therefore divide the remaining exits of `back 1` by $1 - 9/25$ before substituting them in the external clauses.

One call

$$\texttt{exit}\,0 \mapsto \tfrac{3}{5}\texttt{exit}\,1 + \tfrac{4}{5}\texttt{back}\,1 \qquad \texttt{exit}\,1 \mapsto \tfrac{3}{5}\texttt{exit}\,0 + \tfrac{4}{5}\texttt{back}\,0$$
$$\texttt{back}\,0 \mapsto -\tfrac{4}{5}\texttt{exit}\,1 + \tfrac{3}{5}\texttt{back}\,1 \qquad \texttt{back}\,1 \mapsto -\tfrac{4}{5}\texttt{exit}\,0 + \tfrac{3}{5}\texttt{back}\,0$$

Remove `back 0`

$$\texttt{exit}\,1 \mapsto \tfrac{3}{5}\texttt{exit}\,0 + \tfrac{4}{5}\big(-\tfrac{4}{5}\texttt{exit}\,1 + \tfrac{3}{5}\texttt{back}\,1\big) = \tfrac{3}{5}\texttt{exit}\,0 - \tfrac{16}{25}\texttt{exit}\,1 + \tfrac{12}{25}\texttt{back}\,1,$$
$$\texttt{back}\,1 \mapsto -\tfrac{4}{5}\texttt{exit}\,0 + \tfrac{3}{5}\big(-\tfrac{4}{5}\texttt{exit}\,1 + \tfrac{3}{5}\texttt{back}\,1\big) = -\tfrac{4}{5}\texttt{exit}\,0 - \tfrac{12}{25}\texttt{exit}\,1 + \tfrac{9}{25}\texttt{back}\,1.$$

Close `back 1`

$$\texttt{back}\,1 \mapsto \tfrac{25}{16}\big(-\tfrac{4}{5}\texttt{exit}\,0 - \tfrac{12}{25}\texttt{exit}\,1\big) = -\tfrac{5}{4}\texttt{exit}\,0 - \tfrac{3}{4}\texttt{exit}\,1.$$

Both exits

$$\texttt{exit}\,0 \mapsto \tfrac{3}{5}\texttt{exit}\,1 + \tfrac{4}{5}\big(-\tfrac{5}{4}\texttt{exit}\,0 - \tfrac{3}{4}\texttt{exit}\,1\big) = -\texttt{exit}\,0,$$
$$\texttt{exit}\,1 \mapsto \tfrac{3}{5}\texttt{exit}\,0 - \tfrac{16}{25}\texttt{exit}\,1 + \tfrac{12}{25}\big(-\tfrac{5}{4}\texttt{exit}\,0 - \tfrac{3}{4}\texttt{exit}\,1\big) = -\texttt{exit}\,1.$$

Fig. 5. Compiling `trace flipRound`. The top row gives the four clauses of $R_\theta \otimes \mathrm{X}$, read from input to output. Purple marks the return being removed and the expression substituted for it. After the two substitutions, both inputs leave unchanged with amplitude -1.

For a general body, we evaluate each basis input as in Section 2 and choose a return $\texttt{back}\,q$. Write the evaluated clause as $\texttt{back}\,q \mapsto d\,(\texttt{back}\,q) + e$, where d is the self-entry of q in the current table and e contains the other outputs. If $d \neq 1$, we replace each occurrence of $\alpha\,(\texttt{back}\,q)$ in the other clauses by $\alpha e/(1 - d)$, collect equal patterns and delete the clause for q. If $d = 1$, unitarity makes the other entries in its row and column zero, so we can delete it directly, as in `extraPort`.

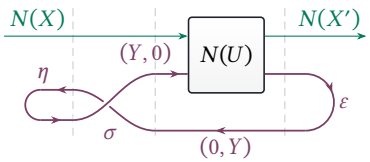
Proposition 3.2 (Compiling away the loop). *Let the evaluated clause table of a finite unitary body have coefficients in a subfield $\mathbb{F} \subseteq \mathbb{C}$ with decidable equality. Eliminating the $|\mathcal{B}(c)|$ return values in any order, one per step, gives a loop-free table over $\mathbb{F}$ denoting L.*

To recover the original wire equations we put back the value we substituted, choosing any value when $d = 1$. Each step preserves unitarity, so we can continue until only the external clauses remain (Appendix B.1). Dropping the outer `exit` tags gives a loop-free table. In `kickbackExact`, for example, eliminating the returns gives $I \otimes X$: the data bit flips even on inputs entangled with the ancilla (Appendix F.1).

Solving the scalar wire equation when $d \neq 1$ gives Mason's signal-flow reduction [36, §§ 3.1–3.2], also used for zero-delay feedback when $I - D$ is invertible [22, Thm. 12, Eq. (12)] and for optical networks [46, App. A, Eq. (A1)]. With n external and m returning values, m steps each update at most $(n+m)^2$ entries. This costs $O(m(n+m)^2)$ field operations, with decidable equality for the test $d = 1$. These dimensions count basis values (2^q for q qubits). Compiling the resulting table into hardware gates is a further step.

Bends and crossings. In Unitary the trace $L = \mathrm{Tr}^Y(U)$ closes the return wire in one step. To draw the two bends separately, we need a type for the wire that runs back. An object $P = (P^+, P^-)$ of Int(Unitary) gives the spaces of rightward and leftward ports. A morphism $P \to Q$ takes input from both ends, so its matrix is a unitary $P^+ \oplus Q^- \to Q^+ \oplus P^-$ [31, § 4]. The embedding N keeps each gate's matrix and sends Y to $(Y, 0)$. Since 0 is the zero-dimensional space, $(Y, 0)$ carries amplitude only to the right and $(0, Y)$ only to the left.

Follow the return output of $N(U)$ around the cap ε. It travels left to the cup η and turns right. The crossing σ swaps the two legs to bring the forward leg to the return input (Appendix Figure 23(b)). Both bends pass the amplitude unchanged, so the return still satisfies $y = Cu + Dy$. The composite is $N(L)$, whose matrix is the gate we just compiled.



3.4 Making the sign observable

The clause for `trace guardRound` sends $|*\rangle$ to $-|*\rangle$, so a measurement gives $*$ with certainty, just as for the identity. To see the sign we need a second amplitude to compare with the one that went through the loop. In `bypassRound`, both belong to one qubit. Figure 6 draws $|0\rangle$ going straight through and $|1\rangle$ passing through `guardRound`, whose return still carries `Unit`.

```
round bypassRound : Qubit + Unit = {
  exit 0 ↔ exit 0;
  exit 1 ↔ [ 3/5 . exit 1 + 4/5 . back unit];
  back unit ↔ [-4/5 . exit 1 + 3/5 . back unit]
}
```

$$U_{\text{bypass}} = \left(\begin{array}{cc|c} 1 & 0 & 0 \\ 0 & t & -s \\ \hline 0 & s & t \end{array}\right)$$

The columns follow the clauses in order, with $t = 3/5$ and $s = 4/5$. The bypass passes straight through with amplitude one and the rotation exits immediately with amplitude t, giving $x_0 = \mathrm{diag}(1, t)$. Every later exit comes from `back unit` and gives $x_{k+1} = \mathrm{diag}(0, -s^2 t^k)$ for $k \geq 0$. Closing the loop leaves $|0\rangle$ alone and sends $|1\rangle$ to $-|1\rangle$.

The caller runs the Hadamard test [13, p. 1, Eqs. (1)–(3)]. In `H; trace bypassRound; H on 0`, the first Hadamard H prepares $(|0\rangle + |1\rangle)/\sqrt{2}$, and the loop changes the plus to a minus. Either state gives equally likely bits if measured now. With exact coefficients $\pm 1/\sqrt{2}$, the second Hadamard

brings the paths together:

$$H\frac{|0\rangle - |1\rangle}{\sqrt{2}} = \tfrac{1}{2}(|0\rangle + |1\rangle) - \tfrac{1}{2}(|0\rangle - |1\rangle) = (\tfrac{1}{2} - \tfrac{1}{2})|0\rangle + (\tfrac{1}{2} + \tfrac{1}{2})|1\rangle = |1\rangle.$$

The program returns $|1\rangle$, so a basis measurement (Section 5) reads 1 with certainty. Had the loop preserved the sign, the two Hadamards would have returned it to 0.

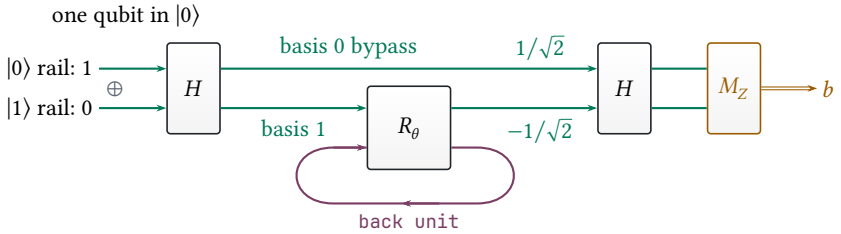


Fig. 6. The two rails carry the basis alternatives of one qubit $G = \mathbb{C}^2$. Each $H : G \to G$ and the closed loop $\mathrm{diag}(1, -1) : G \to G$ are in Unitary. Both rails enter the measurement channel $M_Z : G \rightsquigarrow G$ in CPTP. Its double wire carries one classical bit, here 1 with certainty.

4 The coherent loop in light

Fairbanks and Selinger ask whether a device can take the trace of a unitary supplied as a black box [17, App. B]. For a given matrix, we can build its optical network.

Proposition 4.1 (Optical feedback). *Realize a finite unitary body $U : X \oplus Y \to X' \oplus Y$ at the working frequency as a lossless optical network, one mode per basis value [41]. Connect each returning output to its matching input by a lossless delay of a whole number of wave periods. On steady input u at that frequency the output is Lu.*

Proof. At the working frequency each delay line returns its field unchanged. The steady amplitudes y on the returning inputs and v on the external outputs therefore satisfy $y = Cu + Dy$ and $v = Au + By$, the wire equations of Section 3. By Theorem 3.1, every solution gives $v = Lu$. □

Building the network takes 2^n modes for n qubits. If the return ports are already accessible, we need only add the delays. A fixed number of black-box calls gives a continuous response (Appendix B). In the ring, when little light crosses the gap on each pass, it can take many trips for the output to approach its final amplitude.

The splitters in Figure 7 act like the Hadamards in Figure 6. When the ring reverses the sign, light leaves by the difference port. At $z = -1$ the ring preserves the sign and light leaves by the sum port. Wang et al. used a ring on the shorter path to send alternate frequencies in a regularly spaced sequence mostly to opposite outputs [48, Figs. 1 and 6].

Fairbanks and Selinger point out a difficulty for a device that takes the trace of a unitary. The rotation varies continuously with its angle, yet its trace is $+1$ at $\theta = 0$ and -1 for $0 < \theta < 2\pi$ [17, App. B, Remark B.1]. In the ring, the trace gives the amplitude we approach as light from more and more trips reaches the output. To follow that buildup, suppose we send a steady unit input into an empty ring. We choose the frequency so that one trip takes a whole number of wave cycles. At the coupler, a copy delayed by two trips then lines up crest for crest with one delayed by a single trip. This is resonance. The direct field $t = \cos\theta$ arrives first. After $n \geq 1$ trips the returning light brings the output to $t - (1 - t^2)(1 + t + \cdots + t^{n-1}) = -1 + (1 + t)t^n$.

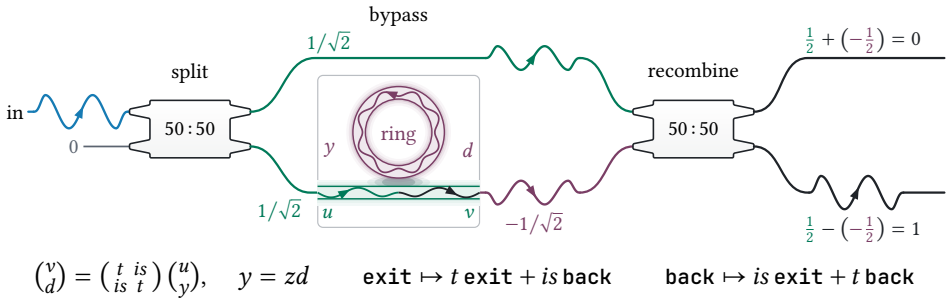
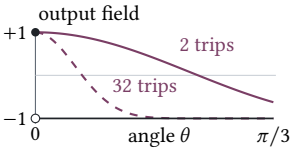


Fig. 7. The second splitter takes the normalized sum of the two paths at the upper port and their normalized difference at the lower. At the gap, light stays in its guide with amplitude $t = 3/5$ or crosses with $is = 4i/5$ [2, 40] (Appendix H). The guide carries u to v and the ring feeds d back as $y = zd$. At $z = 1$ the two crossings give $(is)^2 = -s^2$ and each extra trip contributes t, just as in the delayed exits BD^kCu of `guardRound`. The plot varies $t = \cos\theta$. Black shows the steady output, including the isolated point at +1.



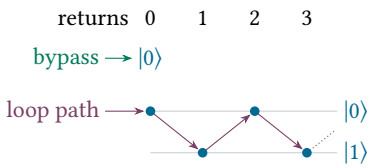
At $\theta = 0$ no light crosses the gap. For any fixed number of trips, the output approaches +1 as the angle approaches zero (Figure 7). If we hold a small positive angle fixed and wait, t^n decays to zero and the output approaches −1.

At resonance, unit input gives a circulating amplitude of magnitude $|s|/(1-t)$ for $s \neq 0$. Its squared magnitude, $s^2/(1-t)^2 = \cot^2(\theta/2)$, grows without bound as $\theta \to 0$. To read this weight, we vary the frequency slightly. If the coupler acts the same at nearby frequencies, the output phase changes with angular frequency at a rate equal to the weight times the time for one trip. This is the Wigner–Smith delay [26, 44, 50].

A flash shorter than one trip behaves differently: the arrival time of each pulse records the count [40, Eqs. (30)–(33)]. In the program we can choose how much of that record to keep.

5 What allocation and hiding do to the loop

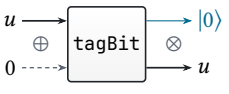
In `bypassRound`, start a bit at 0 and flip it on each return. Exits on calls three and five have made two and four returns, so both carry the same 0 as the bypass. They can still interfere with it. An exit after an odd number of returns carries 1 and is distinguishable from all three.



The caller did not supply a count register. We need the ability to allocate that register.

Adding allocation to ISO. Our caller supplies a qubit with two basis values, but returning it alongside a counter bit gives four output pairs. Tagging the inputs gives a square table in ISO.

```
tagBit :: (Qubit + Qubit) ↔ (Qubit x Qubit)
tagBit = {left x  ↔ ⟨0, x⟩;
          right x ↔ ⟨1, x⟩}
lift tagBit on 1
```



The bit records the input tag, so either clause can run backwards. With `lift` the data enter on the left, leaving zero amplitude on the right. Thus `lift tagBit on 1` gives $\langle 0, 1\rangle$ and on the caller's superposition $u = (|0\rangle + |1\rangle)/\sqrt{2}$ it gives $|0\rangle \otimes u$.

Heunen and Kaarsgaard call their allocation language $\mathrm{UП}_a$ ("yuppie-a") [27, § 3.3]. Our executable language $\textsc{iso}_a$ uses the same subscript. We write arr to embed an iso and lift to allocate.

$$\frac{\omega \,:\, a \leftrightarrow b}{\mathsf{arr}\,\omega \,:\, a \rightarrowtail b} \qquad \frac{\omega \,:\, a + d \leftrightarrow b}{\mathsf{lift}\,\omega \,:\, a \rightarrowtail b}.$$

If ω evaluates to U, then $\mathsf{arr}\,\omega$ uses the whole matrix, while $\mathsf{lift}\,\omega$ gives $U(u, 0)$ on input u. The columns for a are orthonormal, so this restriction still preserves inner products and gives an *isometry*, written $\rightarrowtail$. Two extensions that agree on those columns give the same isometry. This is the identification made by Huot and Staton's completion by allocation [29, Thm. III.3]. Their equivalence, stated by Heunen and Kaarsgaard, identifies the category $\mathscr{R}[\mathsf{Unitary}]$ with Isometry under both sum and tensor [27, Thm. 22]. For $[K]$, the sum of $K \geq 1$ copies of 1, alloc_j prepares j alongside the data. Starting at zero and advancing on each return gives $\mathsf{count}_K\,\beta \,:\, a \rightarrowtail [K] \otimes b$ for $\beta \,:\, a + c \leftrightarrow b + c$. In `count K f on t`, the numeral `K` gives the counter size, `f` the round and `t` its input. In formulas we put the count first to label each exit.

Adding hiding to $\textsc{iso}_a$. For the second Hadamard the caller wants just the data. We need the ability to hide the counter. In `bypassRound`, let u_0 sum the even exits, including the bypass, and u_1 the odd exits. The joint output is $w = |0\rangle \otimes u_0 + |1\rangle \otimes u_1$. To keep just the data, we take the partial trace of $ww^\dagger$ over the bit. Grouping rows and columns by the counter gives

$$\begin{pmatrix} u_0 u_0^\dagger & u_0 u_1^\dagger \\ u_1 u_0^\dagger & u_1 u_1^\dagger \end{pmatrix} \xrightarrow{\text{hide the bit}} u_0 u_0^\dagger + u_1 u_1^\dagger.$$

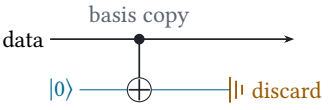
Only the diagonal blocks survive because $\langle 0 \mid 1\rangle = 0$. Within each block the exits carrying that bit have already added their amplitudes.

Heunen and Kaarsgaard add hiding in $\mathrm{UП}_a^\chi$ ("yuppie-chi-a") [27, § 3.4]. Our $\textsc{iso}_a^\chi$ embeds each isometry with arr. We write hide for their second lift to name the output we discard.

$$\frac{v \,:\, a \rightarrowtail b}{\mathsf{arr}\,v \,:\, a \rightsquigarrow b} \qquad \frac{v \,:\, a \rightarrowtail b \otimes g \qquad \mathscr{B}(g) \neq \emptyset}{\mathsf{hide}\,v \,:\, a \rightsquigarrow b}.$$

The rule hides the right factor, so we swap first in $\mathsf{hide}(\mathsf{count}_K\,\beta; \mathsf{arr}\,\sigma_{[K],b})$. Executable `count` returns the data first, so `discard` can hide the counter directly. On a density matrix ρ (a positive operator of trace one), $\mathsf{arr}\,v$ gives $V\rho V^\dagger$ and $\mathsf{hide}\,v$ gives $\mathrm{tr}_g(V\rho V^\dagger)$, where V is the matrix of v. Isometric relabelings of g generate the equivalence in Heunen and Kaarsgaard's $\mathscr{L}[\mathsf{Isometry}]$ [27, Eq. (3), Thm. 22]. On isometries their quotient agrees with Huot and Staton's affine reflection: both give the category CPTP of completely positive trace-preserving maps with composition and tensor [28, Cor. 7]. To discard an inhabited b as in Appendix Figure 14, we hide b in $b \leftrightarrow 1 \otimes b$. This sends ρ to $\mathrm{tr}\,\rho$.

From a fresh bit to a measurement. Start a fresh bit at 0 and flip it when the data bit is 1. On basis input s this writes a copy of s. On a superposition $\alpha|0\rangle + \beta|1\rangle$, the two wires carry $\alpha|00\rangle + \beta|11\rangle$. Discarding the copy destroys the coherence between 0 and 1. Reading the data now gives 0 with probability $|\alpha|^2$ and 1 with probability $|\beta|^2$.



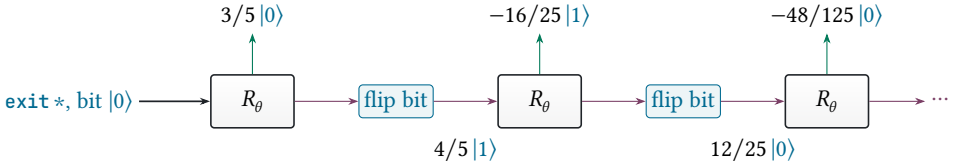
For any inhabited b, we lift a permutation that sends each selected input s to $\langle s, s\rangle$. This gives the basis copier $\mathsf{clone}_b \,:\, b \rightarrowtail b \otimes b$. Hiding its copy measures b in the computational basis [27, § 3.4.1].

$$\mathsf{measure}_b = \mathsf{hide}\,\mathsf{clone}_b, \qquad [\![\mathsf{measure}_b]\!](\rho) = \sum_{s \in \mathscr{B}(b)} P_s \rho P_s, \qquad P_s = |s\rangle\langle s|.$$

The separate construct $\mathsf{while}_c\,\beta \,:\, a \rightsquigarrow b$ copies the tag after each call and discards the copy. Coherence within each payload is preserved. Appendix A.5 gives the rules for empty types.

5.1 Keep a count

In `count 2 guardRound on unit`, the 2 tells us to keep the count modulo two. The first call sends $3/5\,|0\rangle$ to the exit with the bit still at 0. The other $4/5$ goes around with the bit flipped to 1. After two returns the bit is 0 again, so the first and third exits carry the same bit and payload.

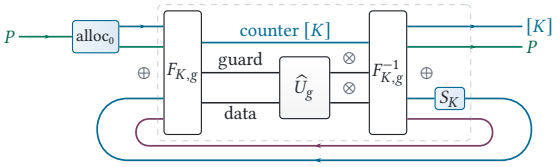


Every two returns multiply a delayed exit by $(3/5)^2$. The even tail sums to $-3/5$, canceling the immediate $3/5$. Only the odd exits survive.

$$\underbrace{\frac{3}{5} - \frac{48/125}{1-(3/5)^2}}_{\text{even returns}} = 0, \qquad \underbrace{\frac{-16/25}{1-(3/5)^2}}_{\text{odd returns}} = -1.$$

The loop path thus leaves with $-|1\rangle$ in the counter and the bypass with $|0\rangle$. On input $|+\rangle$ the joint output is $(|0\rangle \otimes |0\rangle - |1\rangle \otimes |1\rangle)/\sqrt{2}$, with the counter first. Hiding it leaves the data in $I/2$, which the second Hadamard preserves, so the caller reads 1 half the time.

With K values, the bit flip becomes $S_K : j \mapsto j+1 \bmod K$. For a common payload P, $F_{K,g}$ separates counter, guard and payload. The counter passes beside the body. After $F_{K,g}^{-1}$ restores the ports, the return carries both counter and payload. Only the counter passes through S_K.



We allocate the counter at zero and close the return wire of the enlarged body β_K. After k returns the exit $x_k u$ carries $k \bmod K$ (Figure 8):

$$\mathrm{count}_K\,\beta = \mathrm{alloc}_0\,;\,\mathrm{arr}(\mathrm{trace}_{[K]\otimes c}\,\beta_K), \qquad R_K u = \sum_{k\geq 0} |k \bmod K\rangle \otimes x_k u. \tag{5.1}$$

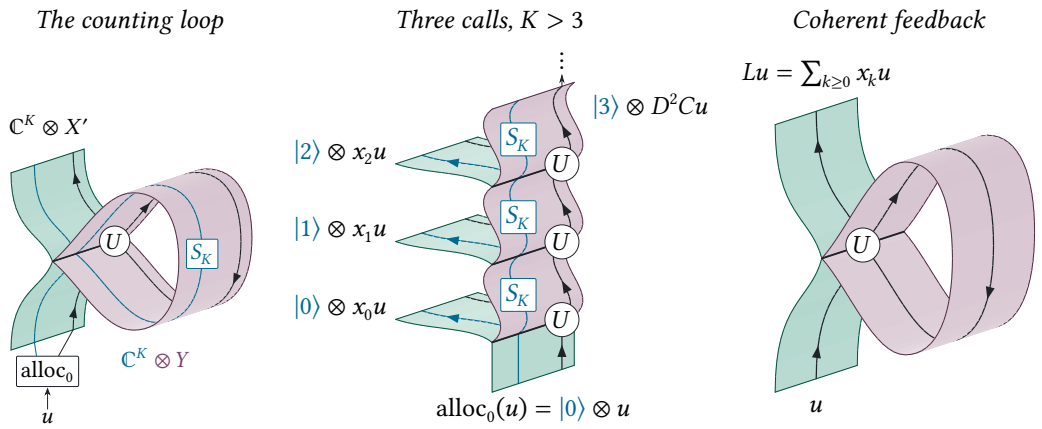


Fig. 8. Read upwards. Each trip around the folded sheet passes through S_K, advancing the counter. After three calls, $|3\rangle \otimes D^2Cu$ continues around the loop. Without the counter, all exit amplitudes add at one output.

Knowing the body, the caller can reverse the enlarged loop to recover $|0\rangle \otimes u$, then apply L to the data. Even without the body, a Fourier measurement can erase the count: the uniform outcome has

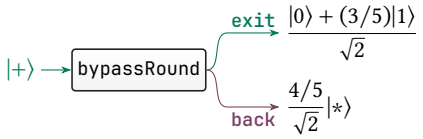
overlap $1/\sqrt{K}$ with every label and yields the unnormalized output $Lu/\sqrt{K}$, with probability $1/K$. When we hide the parity bit, the first and third exits can still interfere because both leave it at zero. For `bypassRound` the probability of 1 approaches $1/5$ as K grows but differs from `while` at every finite K (Appendix C). Allocating the counter and applying the enlarged unitary makes R_K an isometry. The exact record, $Ru = \sum_k |k\rangle \otimes x_k u$ in $\ell^2(\mathbb{N}) \otimes X'$, lies outside the finite language. It too preserves the norm since $\sum_k \|x_k u\|^2 = \|u\|^2$ (Theorem 3.1). Until the counter wraps, its labels agree with the exact ones. The remaining exits have vanishing total norm as K grows, so embedding the first K labels gives $R_K \to R$ in operator norm (Appendix C). A fresh coin at each unfolding can encode the same count [54, § 6]: in Ying's loop, k returns followed by an exit spell $1^k 0$ [52, Eq. (34)]. Copying each guard test into a fresh qubit likewise leaves a record of the outcomes [4, § 4].

In the bounded loop of Yuan et al., the count is already an input giving the number of repetitions. Shorter branches execute no-ops until all reach the same control state at the same step [55, § 6.1, Fig. 7]. This padding assumes a classical upper bound on the number of repetitions. As in constant-time code, padding makes the timing the same for every count. The input retains the original count. Register-machine compilation also computes waits within a time bound [56, § 5]. Our counter records when amplitude leaves, even when one input gives exits at infinitely many counts.

5.2 Measure the returning tag

On the same input `0`, the caller now runs `H; while bypassRound; H; measure`. The counter gave the first and third exits the same bit. Measuring the tag after every call distinguishes the first exit from every later one.

Each path starts with amplitude $1/\sqrt{2}$. On the first exit all of the bypass meets $3/5$ of the lower path. The second Hadamard takes their difference at output 1, giving $(1 - 3/5)/2 = 1/5$ in amplitude, or $1/25$ in probability.



Only the lower path returns, with probability $|(4/5)/\sqrt{2}|^2 = 8/25$. Its chance of returning again is $9/25$ at each call, so it eventually exits as $|1\rangle$. The second Hadamard sends half that probability to output 1, giving $1/25 + (8/25)(1/2) = 1/5$ in all.

Send in a pulse shorter than one trip. Only the direct pulse meets the bypass in Figure 9. The later pulses arrive one trip apart and split equally, giving $1/5$ overall [40, Eqs. (30)–(33)]. This reminds us of Bergson's first book, where he observes that we can hear a bell's rhythm, yet to count its strokes we must keep them distinct [7, pp. 86–87]. Here each trip count has its own arrival time.

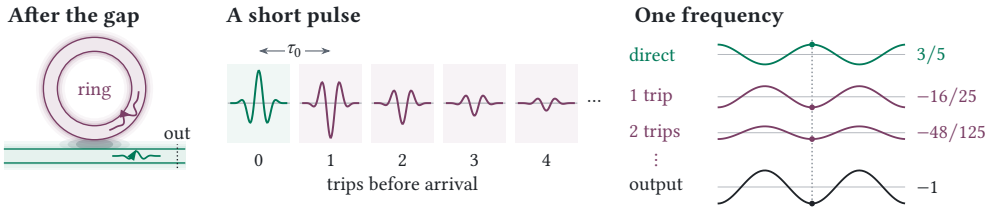


Fig. 9. Successive exits of a short pulse arrive one trip apart. For steady light at $z = 1$, the dots mark the amplitude of each wave at the same instant. They add to give the sign change in Figure 7.

What one call measures. After each call we copy the tag and hide the copy. Both paths carry `exit` on the first exit, so they can still interfere. The body must be reversible. It can contain a compiled `trace`, whereas another `while` has channel type.

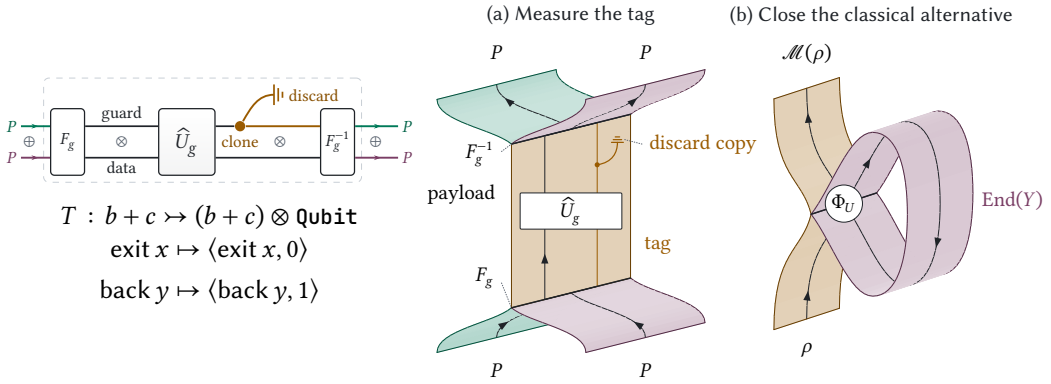


Fig. 10. One call is hide(arr $\beta; T$). With a common payload, F_g exposes the guard for copying and F_g^{-1} restores the ports. The circuit and sheet (a) show one call with both ports open. In (b) the returning density block in End(Y) goes around again.

For input ρ and return σ, we keep the exit and return blocks.

$$\Phi_U(\rho, \sigma) = \big(\underbrace{A\rho A^\dagger + B\sigma B^\dagger}_{\text{exit}}, \underbrace{C\rho C^\dagger + D\sigma D^\dagger}_{\text{return}}\big).$$

For a normalized input the block traces give the two probabilities. On $(\rho, 0)$ the first exit is $A\rho A^\dagger$ and the next is $(BC)\rho(BC)^\dagger$. Ying and Feng test the guard first [53], so when the external types agree we run U once before their loop. Each unfolding then adds one more of our exits, so Ying's least fixed point [51] gives their sum, also the trace over a classical sum [43, § 6.4] (Appendix C).

Adding the measured exits. An exit after k returns gives `exit` on call $k + 1$. Every later exit gives `back` on that call, so the measurement tells them apart. For `guardRound`, the first two exit amplitudes add to $3/5 - 16/25 = -1/25$ coherently. With measurement, their probabilities add to $(3/5)^2 + (-16/25)^2$ (Appendix Table 2).

We can defer measurement by keeping the tag outcomes in orthogonal states. Hiding this exact record adds the exits as density matrices:

$$\mathscr{M}(\rho) = \sum_{k \geq 0} x_k \rho x_k^\dagger = \mathrm{tr}_{\ell^2(\mathbb{N})}(R\rho R^\dagger), \qquad \mathscr{M} = [\![\mathsf{while}_c\ \beta]\!]. \tag{5.2}$$

Since $\sum_k x_k^\dagger x_k = I$ (Theorem 3.1), the probabilities sum to one even without a last call. A smaller register can realize the same channel without storing the count (Appendix C).

Now consider the three loops together (Table 1). Closing the wire lets all exits interfere. A counter preserves interference within each label, while guard measurement separates every round.

Table 1. Erase, keep and measure, in row order, for `bypassRound` on $|+\rangle$ ($t = 3/5$). The first two rows give state vectors and the last a density matrix. Only the data reach the final H.

Loop	*General output*	*Bypass output*	Pr(1) after H
$\mathsf{trace}_c\ \beta : a \leftrightarrow b$	$Lu = \sum_{k \geq 0} x_k u$	$\lvert - \rangle$	1
$\mathsf{count}_K\ \beta : a \rightarrowtail [K] \otimes b$	$R_K u = \sum_{k \geq 0} \lvert k \bmod K \rangle \otimes x_k u$	$\frac{\lvert 0\rangle \otimes \lvert 0\rangle - \lvert 1\rangle \otimes \lvert 1\rangle}{\sqrt{2}}$	$1/2$ $(K = 2)$
$\mathsf{while}_c\ \beta : a \rightsquigarrow b$	$\mathscr{M}(\rho) = \sum_{k \geq 0} x_k \rho x_k^\dagger$	$\frac{1}{2}\begin{pmatrix} 1 & t \\ t & 1 \end{pmatrix}$	$1/5$

The same caller reads 1 with certainty for coherent feedback, with probability $1/2$ when we keep parity, and with probability $1/5$ when we measure on every round.

6 When can a caller forget the difference?

With the bypass, learning that the qubit returned even once tells us which path it took. In `guardRound` every exit carries `unit`, whatever the count. A compiler can replace the measured loop by the coherent gate if both give the caller the same data on every input. We can see when this works by keeping the exact count for a moment.

Suppose each exit x_k applies the same gate L up to a scalar b_k. Keeping the count gives the joint output

$$Ru = \sum_k |k\rangle \otimes x_k u = \Big(\sum_k b_k |k\rangle\Big) \otimes Lu.$$

For `guardRound`, $L = -1$, so factoring out Lu leaves $-\frac{3}{5}|0\rangle + \frac{16}{25}|1\rangle + \frac{48}{125}|2\rangle + \cdots$ in the counter. Knowing how many returns occurred need reveal nothing about the data. Here the whole count state, including its relative phases, is the same for every input, so the caller receives Lu even after discarding the count.

Theorem 6.1 (Agreement on the returned data). *Let $\beta : a + c \leftrightarrow b + c$ be a finite iso with nonempty external input type a. Let $m = |\mathscr{B}(c)|$ be the number of basis values on the return wire. The following are equivalent.*

(1) The loops agree on the returned data, $\mathscr{M}(\rho) = L\rho L^\dagger$ for every input state ρ.
(2) One unit count vector $\xi \in \ell^2(\mathbb{N})$ satisfies $Ru = \xi \otimes Lu$ for every input u.
(3) The first $m + 1$ exits span a line. One is nonzero and all the others in the list are its scalar multiples. In this case $x_k = b_k L$ for every k, where $\xi = \sum_k b_k |k\rangle$, $\sum_k b_k = 1$ and $\sum_k |b_k|^2 = 1$.

If the loops agree, any direction orthogonal to Lu has probability zero on unit input u. Each exit contributes a nonnegative probability there, so $x_k u$ must lie along Lu. Since L sends distinct basis inputs to orthogonal outputs, a single multiplier on their sum must match both coefficients. The same b_k therefore works for every basis input and, by linearity, $x_k = b_k L$. For finitely many exits this also follows from the freedom in a Kraus representation [49, Cor. 2.23]. Conversely, tracing out a fixed unit count vector leaves $L\rho L^\dagger$.

The immediate exit is A. For $m > 0$, Cayley–Hamilton expresses D^j for every $j \geq m$ in terms of $I, D, \ldots, D^{m-1}$. Multiplying on the left by B and on the right by C puts every later exit BD^jC in the span of $x_1, \ldots, x_m$. Thus $x_0, \ldots, x_m$ span all exits even without a last round. For $m = 0$ only A remains. This finite span is closed, so it contains the sum $L \neq 0$. If the span is a line, every exit is a multiple $b_k L$. The exits sum to L, so $\sum_k b_k = 1$. Their probabilities on a unit input sum to $\sum_k |b_k|^2 = 1$.

For dense matrices with $n = \dim X = \dim X'$, start with $P = C$. Forming BP and updating $P \leftarrow DP$ for successive exits costs $O(m^2n^2 + m^3n)$ field operations. Choose a nonzero entry of the first nonzero exit to fix each multiplier. Checking the other entries costs $O((m + 1)n^2)$ operations and comparisons. This check needs exact arithmetic: near $t = 1$, rounding a small exit to zero can change the bypass caller's probability from $(1 - t)/2$ to one if the replacement is accepted. A direct channel test solves for up to m^2 return entries (Appendix D).

The last exit can matter to the span test. Suppose $|0\rangle$ exits at once and $|1\rangle$ after $m > 0$ returns, both unchanged (Appendix D). A count modulo m labels both exits zero and preserves $|+\rangle$, while measurement leaves $I/2$. All exits between $x_0 = |0\rangle\langle 0|$ and $x_m = |1\rangle\langle 1|$ are zero, so only x_m reveals the second direction.

Replacing the loop in a program. The two paths in Figure 11 can give different data on the same body, as Andrés-Martínez shows in comparing the traces and studying weak guard measurements [3, §§ 3.3.3, 4.2]. The exit maps tell a compiler which bodies allow the replacement. In `flipRound`, the

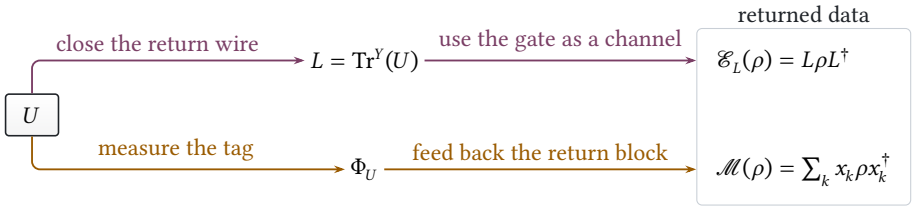


Fig. 11. Closing $U : X \oplus Y \to X' \oplus Y$ in Unitary gives $L : X \to X'$. Measuring the tag gives $\Phi_U : \mathrm{End}(X) \oplus \mathrm{End}(Y) \to \mathrm{End}(X') \oplus \mathrm{End}(Y)$, whose return we close by the classical-sum trace in CPTR (Appendix C). Both routes give the caller a channel from X to X' in CPTP.

first two exits are $(3/5)\mathsf{X}$ and $-(16/25)I$, so they span two directions. In `flipOnExit`, we flip the data only as it leaves. With $t = 3/5$ and $s = 4/5$, the immediate exit is $t\mathsf{X}$. After $k+1$ returns the exit is $-s^2t^k\mathsf{X}$, still a multiple of the same flip.

Leaving a hole in the caller where it uses the loop's data channel gives a *channel context*. We allow any channel before, after or beside the hole, and any finite measurement with the hole on one outcome. Equal channels agree in all these contexts, even on inputs entangled with a reference. If they differ, some input and measurement distinguish them.

Corollary 6.2 (Replacement in a caller). *Let β be a finite iso with m basis values on its return wire. Taking the partial trace over the exact count in $\ell^2(\mathbb{N})$ gives the measured loop. On a nonempty external input type, the compiled coherent gate may replace the measured loop in every channel context if and only if $x_0, \dots, x_m$ span a line. In that case, hiding the counter in* $\mathrm{count}_K\ \beta$ *gives the same channel for every $K \geq 1$. On an empty external input type all these channels agree.*

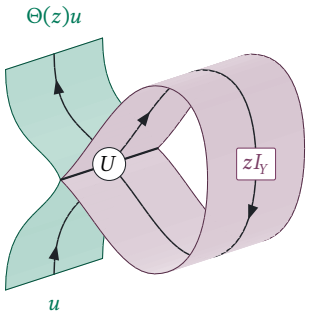
A channel context can keep a reference alongside the closed loop by tensoring its channel with the identity. In `bypassRound` we give the input another path by adding an identity summand to the body *before* closing it. Only the immediate exit acts on that summand. Since every later exit is zero there, it can be a multiple of the immediate exit only if it vanishes everywhere. With an identity bypass, replacement therefore holds exactly when every external input exits on the first call.

7 Reading the returns with a phase

Once `guardRound` is closed, the caller can see the sign change by using a bypass. What the sign alone cannot tell us is how it arose. A clause that exits at once with amplitude -1 gives exactly the same gate as the rotation with feedback. To learn which body we have, we put a phase on the return wire before we close it. Only the rotation sends amplitude through that wire.

In `trace (guardRound; phaseBack)` we run `phaseBack` after every call. Its exit clause passes `exit unit` straight through. Its return clause multiplies `back unit` by $z = e^{i\varphi}$, as in `back unit <-> -1 . back unit` for $z = -1$. A phase placed on the returning state marks each iteration in the amplitude itself. On the purple sheet the direct $3/5$ misses the gate, the first return picks up z and gives $-(16/25)z$, and the next gives $-(48/125)z^2$. With $t = 3/5$ and $s = 4/5$, each further trip contributes another factor tz, giving

$$\Theta_{\mathrm{rot}}(z) = t - s^2 z(1 + tz + t^2z^2 + \cdots) = \frac{t - z}{1 - tz}.$$



On `exit unit` the first call sends amplitude $s = 4/5$ back, so a guard measurement gives `back` with probability $|s|^2 = 16/25$. The next call then starts on `back unit`, where its return amplitude is

$t = 3/5$ and the probability of another return is $|t|^2 = 9/25$. The probabilities of making at least one, two and three returns are s^2, s^2t^2, s^2t^4. Summing these tails counts a run with k returns k times, so the mean is $s^2/(1-t^2) = 1$ whenever $s \neq 0$, since $s^2 = 1 - t^2$.

When we close the wire coherently, the same returns contribute amplitudes $s, stz, s(tz)^2, \ldots$ just before the phase gate. At $z = 1$ they add to $s/(1-t) = 2$ on unit input. The squared magnitude is the *circulating weight*, here $|2|^2 = 4$, four times the measured mean. At $z = -1$ the signs alternate, giving amplitude $s/(1+t) = 1/2$ and weight $1/4$.

The output reveals this weight when we compare nearby phases. Since $\Theta_{\mathrm{rot}}(e^{i\delta}) = -1 - 4i\delta + O(\delta^2)$, its phase changes by 4δ to first order. Heebner and Boyd obtain the same factor for a lossless ring at resonance ($z = 1$): it is the intensity inside the ring divided by the incident intensity [26, p. 848, Eq. (7)]. A trip of duration τ_0 adds phase $\omega\tau_0$ at angular frequency ω, so if the coupler acts the same at nearby frequencies, the delay relative to the direct path is $4\tau_0$ [44, 50].

From one return wire to a finite body. Each trip through the phase gate and D adds one term to the return vector $F_z u = Cu + zDCu + z^2D^2Cu + \cdots$. These returns lie in Y_1, the orthogonal complement of the states that never emit. There $D_1 = D|_{Y_1}$ has geometrically decaying powers (Theorem 3.1), so $F_z = (I_{Y_1} - zD_1)^{-1}C$. Each return crosses the phase gate once more on its way out through B.

Theorem 7.1 (The measured mean and the phase derivative). *For a finite unitary body, let x_k be the exit map after k returns. At $z = e^{i\varphi}$ the coherent output map is $\Theta(z) = A + zBF_z = \sum_{k\geq 0} z^k x_k$. Measuring the guard gives exit k with probability $\|x_k u\|^2$ on unit input u, so the mean number of returns is $\langle u, Nu\rangle$ for $N = \sum_{k\geq 0} k\, x_k^\dagger x_k$. Then*

$$Q(z) = -i\Theta(z)^\dagger \frac{d}{d\varphi}\Theta(e^{i\varphi}) = F_z^\dagger F_z, \qquad \frac{1}{2\pi}\int_0^{2\pi} Q(e^{i\varphi})\, d\varphi = N, \qquad \operatorname{tr} N = \dim Y_1.$$

To see why the output reveals the weight, compare two nearby phases z and $w = ze^{i\delta}$ on the same unit input u. At z the body sends $(u, zF_z u)$ to $(\Theta(z)u, F_z u)$, while the second setting feeds $wF_w u$ back. Equating the input and output inner products gives [6, Eq. (4.10)]

$$\langle \Theta(z)u, \Theta(w)u\rangle = 1 + (e^{i\delta} - 1)\langle F_z u, F_w u\rangle.$$

As w approaches z, the return vectors agree in the limit, so the imaginary part of the overlap is $\delta\|F_z u\|^2$ to first order. Its slope $\langle u, Q(z)u\rangle$ measures the circulating weight. For a fixed return count k, the output simply acquires z^k, so $\Theta(z) = z^k L$ and $Q(z) = kI$.

Averaging $\|F_z u\|^2$ over a full turn cancels the cross terms, leaving $\|Cu\|^2 + \|D_1 Cu\|^2 + \|D_1^2 Cu\|^2 + \cdots$. These are the probabilities of at least one return, at least two, and so on. Their sum counts a run with k returns k times, just as for `guardRound`.

Summing over an orthonormal input basis gives $\operatorname{tr} N$. Unitarity gives $CC^\dagger = I_{Y_1} - D_1 D_1^\dagger$, so

$$\operatorname{tr} N = \sum_{j\geq 0} \operatorname{tr}(D_1^j CC^\dagger D_1^{\dagger j}) = \sum_{j\geq 0} \operatorname{tr}(D_1^j D_1^{\dagger j} - D_1^{j+1} D_1^{\dagger(j+1)}) = \dim Y_1.$$

because consecutive terms cancel and D_1^j tends to zero (Appendix E). For $X' = X$, counting the initial call in a monitored first return gives $\operatorname{tr}(I_X + N) = \dim X + \dim Y_1$ [9, Thm. 3.2, Eq. (9)].

Reading the weight with one qubit. We let a qubit turn a little each time amplitude crosses the return wire. The idea comes from the Larmor clock, which probes the time a particle spends in a region by letting a weak magnetic field turn its spin only there [11, 45].

We prepare the qubit in $(|0\rangle + |1\rangle)/\sqrt{2}$ (Figure 12). On each return the $|0\rangle$ branch picks up z and the $|1\rangle$ branch picks up $ze^{i\delta}$, so after k returns the extra phase on $|1\rangle$ is $e^{ik\delta}$. After the exits have

added, $S^\dagger$ multiplies the second branch by $-i$, so the final Hadamard reads the imaginary part of the overlap [13, p. 1, following Eq. (3)]:

$$\Pr(b=0) - \Pr(b=1) = \operatorname{Im}\langle \Theta(z)u, \Theta(ze^{i\delta})u\rangle = \delta \, \|F_z u\|^2 + O(\delta^2).$$

An immediate exit gives both bits equal probability. For `guardRound` at $z = 1$, the overlap gives a difference of $4\delta + O(\delta^2)$. In n independent runs, the difference in bit frequencies divided by δ estimates four with bias $O(\delta)$ and variance at most $1/(n\delta^2)$ (Appendix I).

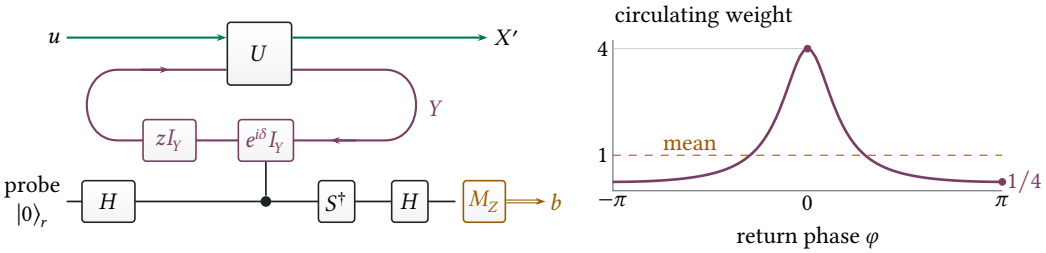


Fig. 12. Before the final gates the state is $(|0\rangle \otimes \Theta(z)u + |1\rangle \otimes \Theta(ze^{i\delta})u)/\sqrt{2}$. The curve gives the weight for `guardRound` and the dashed line its phase average.

Averaging over the phase. Keep δ fixed and choose z uniformly for each run. Averaging z^{k-j} leaves only pairs with equal counts. Each surviving count k turns the probe by $k\delta$, so for `guardRound` the bit probabilities now differ by $\delta + O(\delta^2)$: the probe reads the measured mean of one. The same average on the data gives

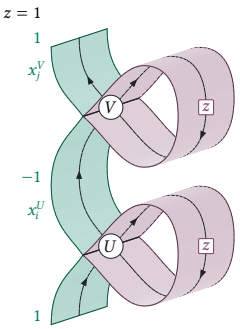
$$\mathcal{M}(\rho) = \frac{1}{2\pi}\int_0^{2\pi} \Theta(e^{i\varphi})\rho\Theta(e^{i\varphi})^\dagger \, d\varphi. \tag{7.1}$$

Without the probe, the bypass caller reads 1 at $z = 1$ and 0 at $z = -1$. Choosing between them by a fair coin gives 1 half the time. Only odd counts change sign, so exits of the same parity still interfere, as with a hidden parity bit (Appendix C).

To average over a full turn, follow the same two paths. The first Hadamard gives each amplitude $1/\sqrt{2}$. At output 1 we subtract the loop path from the bypass and divide by $\sqrt{2}$ again, giving $(1 - \Theta_{\mathrm{rot}}(z))/2$. The loop changes only the phase, so $|\Theta_{\mathrm{rot}}(z)| = 1$ and squaring the magnitude gives probability $(1 - \operatorname{Re}\Theta_{\mathrm{rot}}(z))/2$. With z chosen uniformly, each delayed exit carries a power z^k that averages to zero. Only the immediate $3/5$ remains, giving $(1 - 3/5)/2 = 1/5$.

The bypass reveals a phase that is invisible on `guardRound` alone. If all exits are multiples of one gate, changing z gives that gate an overall phase, leaving its data channel unchanged. Conversely, a channel that stays the same over the circle agrees at $z = 1$ with its average: the coherent and measured loops then give the same data (Appendix E).

8 What the caller can tell



At $z = 1$ each closed `guardRound` changes the sign, so two in sequence look like a straight wire. We put the same probe on both return wires, so a return in either loop turns it by δ.

Follow the green sheet up from U to V. An exit after i returns in U has map x_i^U. Its output enters V, where j further returns give the composite $x_j^V x_i^U$. Each return picks up z, so one trip below and two above carry the same z^3 as three below. Adding all exits with the same total gives $x_k = \sum_{i+j=k} x_j^V x_i^U$. For two copies of the rotation the response is $\Theta_{\mathrm{rot}}(z)^2$.

In two identical rings the arrival time of a short pulse tells us the total number of trips. The direct pulse passes through both rings. One round-trip time later, light that went around either ring arrives in the same pulse.

At the frequency where one trip takes a whole number of cycles, each ring changes the sign and the two signs cancel. Varying the common phase now distinguishes the pair from a straight wire. The two responses multiply, so their phase shifts add. Each contributes 4δ to first order, giving a difference of $8\delta + O(\delta^2)$ in the probe's bit probabilities instead of zero. Averaging over the common phase gives two returns on average, even though the probe still leaves the two ways of making one trip free to interfere.

9 Related work and future directions

At the frequency where the two sign changes cancel, a straight wire gives the same gate as the two rings. Clément and Perdrix redraw polarizing beam-splitter diagrams on at least two wires with invertible gates, removing feedback while preserving the external map [14, § 5, Prop. 26]. Each path there uses a feedback wire at most twice. In `guardRound` arbitrarily many trips contribute to one clause, and Andrés-Martínez's construction lets us sum them one port at a time [3, Thm. 3.3.15].

Which trips can contribute together depends on what survives each call. In Assolini and Di Pierro's loop a fresh qubit records each test, so exits on calls three and five remain distinct [4, Fig. 2 and § 4]. Andrés-Martínez varies how well a measurement distinguishes the guard values [3, § 4.2]. Our parity bit flips on every return, yet both exits leave it at zero.

Future work. Separate counters record where each return took place. A shared counter keeps only the total. We can decide whether sharing is safe by comparing the two data channels (Appendix D). We would like to detect safe sharing from the clauses without building both channels. A similar question arises in recursion, where calls at the same depth can arrive by different routes.

10 Conclusion

We started with three choices for one loop body. Closing the wire lets us add all the exits into one gate. Keeping a count or measuring the guard changes which exits can still interfere, and our test tells us when the caller gets the coherent output even after we discard the exact count. Toss a fair coin before each run and change the sign on every return if it lands heads. Exits two returns apart still add in either case, and averaging the two settings gives the same data as hiding the parity bit. Averaging uniformly over all return phases gives the measured data, and reading the probe then recovers the mean number of returns.

Suppose we already know the total. Let one loop pass $|0\rangle$ straight through and delay $|1\rangle$ by one return, and let the next do the reverse. On input $|+\rangle$ each path makes one return, so a shared counter ends in $|1\rangle$ and leaves the qubit in $|+\rangle$. Separate counters leave $(0, 1)$ beside $|0\rangle$ and $(1, 0)$ beside $|1\rangle$. Discarding them leaves an equal mixture, which the final Hadamard preserves, whereas it sends $|+\rangle$ to $|0\rangle$ (Appendix F). With one counter that bit is always zero. With two counters it is random.

We knew all along that there was one return. What changed the qubit was writing down where it happened. With a shared counter the two paths could still interfere, precisely because knowing the total left that part of the history open.

AI Use. The authors used ChatGPT Codex (GPT-6.1 Sol) for

- formatting, revising prose and generating TeX for diagrams and figures,
- surveying the literature and finding relevant papers, especially in optics,
- implementing allocation and hiding in the software artifact, and
- exploring ideas and checking the reasoning behind them.

The authors verified the results and take responsibility for the paper.

Data-Availability Statement

The software accompanying this paper includes the ISO archive (version 0.1.0.0, SHA-256 prefix `020e6ee2`) and a companion comparison script. It provides three things:

(1) The floating-point interpreter and runnable ISO programs for the examples.
(2) An exact checker for real rational clause tables that compiles coherent loops and tests whether measuring the guard gives the same data.
(3) A comparison on nine bodies that checks both elimination orders against the wire equations and the agreement test against a direct channel calculation.

If the paper is accepted, we plan to submit the artifact for evaluation.

References

[1] Samson Abramsky, Esfandiar Haghverdi, and Philip Scott. 2002. Geometry of Interaction and linear combinatory algebras. *Mathematical Structures in Computer Science* 12, 5 (2002), 625–665. doi:10.1017/S0960129502003730

[2] Paul M. Alsing, Edwin E. Hach, III, Christopher C. Tison, and A. Matthew Smith. 2017. Quantum-optical description of losses in ring resonators based on field-operator transformations. *Physical Review A* 95 (2017), 053828. arXiv:1608.01280 [quant-ph] doi:10.1103/PhysRevA.95.053828

[3] Pablo Andrés-Martínez. 2022. *Unbounded loops in quantum programs: categories and weak while loops.* Ph. D. Dissertation. University of Edinburgh. arXiv:2212.05371 doi:10.48550/arXiv.2212.05371

[4] Nicola Assolini and Alessandra Di Pierro. 2025. A Denotational Semantics for Quantum Loops. arXiv:2506.23320 [cs.PL]. doi:10.48550/arXiv.2506.23320

[5] Miklós Bartha. 2014. Quantum Turing automata. In *Proceedings of DCM 2012 (EPTCS, Vol. 143)*. Open Publishing Association, Waterloo, NSW, Australia, 17–31. arXiv:1404.0074 doi:10.4204/EPTCS.143.2

[6] Jussi Behrndt, Seppo Hassi, and Henk de Snoo. 2009. Boundary Relations, Unitary Colligations, and Functional Models. *Complex Analysis and Operator Theory* 3, 1 (2009), 57–98. doi:10.1007/s11785-008-0064-z

[7] Henri Bergson. 1910. *Time and Free Will: An Essay on the Immediate Data of Consciousness.* Translated by F. L. Pogson. Swan Sonnenschein, London.

[8] Niels Bohr. 1949. Discussion with Einstein on epistemological problems in atomic physics. In *Albert Einstein: Philosopher-Scientist*, Paul Arthur Schilpp (Ed.). The Library of Living Philosophers, Evanston, IL, 199–241.

[9] J. Bourgain, F. A. Grünbaum, L. Velázquez, and J. Wilkening. 2014. Quantum Recurrence of a Subspace and Operator-Valued Schur Functions. *Communications in Mathematical Physics* 329, 3 (2014), 1031–1067. arXiv:1302.7286 doi:10.1007/s00220-014-1929-9

[10] M. S. Brodskii. 1978. Unitary operator colligations and their characteristic functions. *Russian Mathematical Surveys* 33, 4 (1978), 159–191. doi:10.1070/RM1978v033n04ABEH002495.

[11] M. Büttiker. 1983. Larmor precession and the traversal time for tunneling. *Phys. Rev. B* 27, 6178. doi:10.1103/PhysRevB.27.6178.

[12] Jacques Carette, Chris Heunen, Robin Kaarsgaard, and Amr Sabry. 2024. How to Bake a Quantum Π. *Proceedings of the ACM on Programming Languages* 8, ICFP, Article 236 (2024), 29 pages. doi:10.1145/3674625

[13] Andrew M. Childs. 2013. Quantum Algorithms, Lecture 18: Approximating the Jones Polynomial. University of Waterloo lecture notes. https://www.cs.umd.edu/~amchilds/teaching/w13/l18.pdf

[14] Alexandre Clément and Simon Perdrix. 2020. PBS-Calculus: A Graphical Language for Coherent Control of Quantum Computations. In *45th International Symposium on Mathematical Foundations of Computer Science (LIPIcs, Vol. 170)*. Schloss Dagstuhl, Dagstuhl, Germany, 24:1–24:14. doi:10.4230/LIPIcs.MFCS.2020.24

[15] Cole Comfort, Antonin Delpeuch, and Jules Hedges. 2020. Sheet diagrams for bimonoidal categories. arXiv:2010.13361 doi:10.48550/arXiv.2010.13361

[16] Marek Czachor. 2019. Time travel without paradoxes: Ring resonator as a universal paradigm for looped quantum evolutions. *Physics Letters A* 383 (2019), 2704–2712. arXiv:1805.12129v8. doi:10.1016/j.physleta.2019.05.043

[17] Aaron David Fairbanks and Peter Selinger. 2025. On traces in categories of contractions. In *Proceedings of the 22nd International Conference on Quantum Physics and Logic (QPL 2025) (EPTCS, Vol. 426)*, Alejandro Díaz-Caro, Ognyan Oreshkov, and Ana Belén Sainz (Eds.). Open Publishing Association, Waterloo, NSW, Australia, 79–99. arXiv:2502.14993 doi:10.4204/EPTCS.426.3

[18] Y. V. Fyodorov and H.-J. Sommers. 1997. Statistics of resonance poles, phase shifts and time delays in quantum chaotic scattering: Random matrix approach for systems with broken time-reversal invariance. *Journal of Mathematical Physics* 38, 1918–1981. doi:10.1063/1.531919.

[19] Jean-Yves Girard. 1995. Geometry of interaction III: accommodating the additives. In *Advances in Linear Logic*. Cambridge University Press, 329–389. doi:10.1017/CBO9780511629150.017.

[20] Robert Glück and Robin Kaarsgaard. 2018. A Categorical Foundation for Structured Reversible Flowchart Languages: Soundness and Adequacy. *Logical Methods in Computer Science* 14, 3, Article 16 (2018), 38 pages. doi:10.23638/LMCS-14(3:16)2018 (arXiv:1710.03666).

[21] John E. Gough. 2014. Feedback network models for quantum transport. *Physical Review E* 90 (2014), Article 062109. doi:10.1103/PhysRevE.90.062109

[22] John Gough and Matthew R. James. 2009. Quantum Feedback Networks: Hamiltonian Formulation. *Communications in Mathematical Physics* 287 (2009), 1109–1132. doi:10.1007/s00220-008-0698-8

[23] F. A. Grünbaum, L. Velázquez, A. H. Werner and R. F. Werner. 2013. Recurrence for Discrete Time Unitary Evolutions. *Communications in Mathematical Physics* 320, 543–569. doi:10.1007/s00220-012-1645-2.

[24] Masahito Hasegawa. 1997. Recursion from Cyclic Sharing: Traced Monoidal Categories and Models of Cyclic Lambda Calculi. In *Typed Lambda Calculi and Applications (Lecture Notes in Computer Science, Vol. 1210)*. Springer, 196–213. doi:10.1007/3-540-62688-3_37

[25] S. W. Hawking. 1992. Chronology protection conjecture. *Physical Review D* 46, 2 (1992), 603–611. doi:10.1103/PhysRevD.46.603.

[26] John E. Heebner and Robert W. Boyd. 1999. Enhanced all-optical switching by use of a nonlinear fiber ring resonator. *Optics Letters* 24, 12, 847–849. doi:10.1364/OL.24.000847.

[27] Chris Heunen and Robin Kaarsgaard. 2022. Quantum information effects. *Proc. ACM Program. Lang.* 6, POPL (2022), 2:1–2:27. arXiv:2107.12144 doi:10.1145/3498663

[28] Mathieu Huot and Sam Staton. 2019. Universal Properties in Quantum Theory. In *Proceedings of the 15th International Conference on Quantum Physics and Logic (QPL 2018)*, EPTCS 287, 213–223. doi:10.4204/EPTCS.287.12

[29] Mathieu Huot and Sam Staton. 2019. Quantum Channels as a Categorical Completion. In *34th Annual ACM/IEEE Symposium on Logic in Computer Science (LICS)*, 1–13. doi:10.1109/LICS.2019.8785700

[30] Roshan P. James and Amr Sabry. 2013. Isomorphic Interpreters from Logically Reversible Abstract Machines. In *Reversible Computation: 4th International Workshop, RC 2012, Revised Papers (Lecture Notes in Computer Science, Vol. 7581)*. Springer, Berlin, Heidelberg, 57–71. doi:10.1007/978-3-642-36315-3_5

[31] André Joyal, Ross Street, and Dominic Verity. 1996. Traced monoidal categories. *Mathematical Proceedings of the Cambridge Philosophical Society* 119, 3 (1996), 447–468. doi:10.1017/S0305004100074338

[32] Robin Kaarsgaard and Mathys Rennela. 2021. Join Inverse Rig Categories for Reversible Functional Programming, and Beyond. In *Proceedings of the 37th Conference on the Mathematical Foundations of Programming Semantics (MFPS 2021) (EPTCS, Vol. 351)*. Open Publishing Association, Waterloo, NSW, Australia, 152–167. doi:10.4204/EPTCS.351.10 (arXiv:2105.09929).

[33] Louis Lemonnier. 2024. *The Semantics of Effects: Centrality, Quantum Control and Reversible Recursion.* Ph. D. Dissertation. Université Paris-Saclay. arXiv:2406.07216 doi:10.48550/arXiv.2406.07216

[34] Noah Linden and Sandu Popescu. 1998. The halting problem for quantum computers. arXiv:quant-ph/9806054 doi:10.48550/arXiv.quant-ph/9806054

[35] Octavio Malherbe, Philip Scott, and Peter Selinger. 2012. Partially traced categories. *Journal of Pure and Applied Algebra* 216, 12 (2012), 2563–2585. arXiv:1107.3608 doi:10.1016/j.jpaa.2012.03.026

[36] Samuel J. Mason. 1953. Feedback Theory: Some Properties of Signal Flow Graphs. Technical Report 153. Research Laboratory of Electronics, MIT.

[37] Microsoft. 2025. Conditional loops in Q#. *Microsoft Learn.* https://learn.microsoft.com/en-us/azure/quantum/user-guide/language/expressions/conditionalloops.

[38] John M. Myers. 1997. Can a Universal Quantum Computer Be Fully Quantum? *Physical Review Letters* 78, 1823–1824. doi:10.1103/PhysRevLett.78.1823.

[39] OpenQASM. Classical instructions. *OpenQASM Language Specification*, section on while loops. Accessed November 8, 2026. https://openqasm.com/language/classical.html#while-loops.

[40] M. G. Raymer and C. J. McKinstrie. 2013. Quantum Input-Output Theory for Optical Cavities with Arbitrary Coupling Strength: Application to Two-Photon Wave-Packet Shaping. *Physical Review A* 88 (2013), 043819. arXiv:1309.5915 [quant-ph] doi:10.1103/PhysRevA.88.043819

[41] Michael Reck, Anton Zeilinger, Herbert J. Bernstein, and Philip Bertani. 1994. Experimental realization of any discrete unitary operator. *Physical Review Letters* 73 (1994), 58–61. doi:10.1103/PhysRevLett.73.58

[42] Amr Sabry, Benoît Valiron, and Juliana Kaizer Vizzotto. 2018. From symmetric pattern-matching to quantum control. In *FoSSaCS 2018*. Springer, Cham, 348–364. arXiv:1804.00952 doi:10.1007/978-3-319-89366-2_19

[43] Peter Selinger. 2004. Towards a quantum programming language. *Mathematical Structures in Computer Science* 14, 4, 527–586. doi:10.1017/S0960129504004256.

[44] Felix T. Smith. 1960. Lifetime Matrix in Collision Theory. *Physical Review* 118, 1 (1960), 349–356. doi:10.1103/PhysRev.118.349

[45] Aephraim M. Steinberg. 1995. How much time does a tunneling particle spend in the barrier region? *Phys. Rev. Lett.* 74, 2405–2409. doi:10.1103/PhysRevLett.74.2405.

[46] Nikolas Tezak, Armand Niederberger, Dmitri S. Pavlichin, Gopal Sarma, and Hideo Mabuchi. 2012. Specification of photonic circuits using Quantum Hardware Description Language. *Philosophical Transactions of the Royal Society A* 370, 5270–5290. doi:10.1098/rsta.2011.0526.

[47] Matt Visser. 1993. From wormhole to time machine: Remarks on Hawking's chronology protection conjecture. *Physical Review D* 47, 2 (1993), 554–565. doi:10.1103/PhysRevD.47.554.

[48] Songli Wang, Yuyang Wang, Swarnava Sanyal, Robert Parsons, Xingchen Ji, Yoshitomo Okawachi, Xiang Meng, Michal Lipson, Alexander Gaeta, and Keren Bergman. 2025. Automated tuning of a ring-assisted MZI-based interleaver for Kerr frequency combs. *Optics Letters* 50, 2 (2025), 698–701. doi:10.1364/OL.546772

[49] John Watrous. 2018. *The Theory of Quantum Information.* Cambridge University Press, Cambridge, UK. doi:10.1017/9781316848142

[50] Eugene P. Wigner. 1955. Lower Limit for the Energy Derivative of the Scattering Phase Shift. *Physical Review* 98, 145–147. doi:10.1103/PhysRev.98.145.

[51] Mingsheng Ying. 2011. Floyd–Hoare Logic for Quantum Programs. *ACM Transactions on Programming Languages and Systems* 33, 6, Article 19 (2011), 49 pages. arXiv:0906.4586v1. doi:10.1145/2049706.2049708

[52] Mingsheng Ying. 2014. Quantum Recursion and Second Quantisation. arXiv:1405.4443 [quant-ph]. doi:10.48550/arXiv.1405.4443

[53] Mingsheng Ying and Yuan Feng. 2010. Quantum loop programs. *Acta Informatica* 47, 4 (2010), 221–250. arXiv:quant-ph/0605218 doi:10.1007/s00236-010-0117-4

[54] Mingsheng Ying, Nengkun Yu, and Yuan Feng. 2012. Defining Quantum Control Flow. arXiv:1209.4379v1.

[55] Charles Yuan, Agnes Villanyi, and Michael Carbin. 2024. Quantum control machine: the limits of control flow in quantum programming. *Proc. ACM Program. Lang.* 8, OOPSLA1 (2024), 1–28. arXiv:2304.15000 doi:10.1145/3649811

[56] Zhicheng Zhang and Mingsheng Ying. 2025. Quantum register machine: efficient implementation of quantum recursive programs. *Proc. ACM Program. Lang.* 9, PLDI, Article 180 (2025), 26 pages. arXiv:2408.10054 doi:10.1145/3729283

A Typing and interpreting the clauses

A.1 Syntax

In `trace guardRound on unit`, `trace guardRound` closes the return wire of the named table and `on` applies the resulting iso to `unit`. Figure 13 gives the grammar. Types are built from 0 and 1 by sum and tensor, written `+` and `x` in source types. In programs, `;` denotes sequence, `+` direct sum, `*` tensor and `dagger` the adjoint.

$$
\begin{aligned}
a, b &::= 0 \mid 1 \mid a + b \mid a \otimes b \\
v &::= * \mid x \mid \mathsf{left}\ v \mid \mathsf{right}\ v \mid \langle v_1, v_2 \rangle \mid \mathsf{exit}\ v \mid \mathsf{back}\ v \\
p, q &::= * \mid x \mid \langle p, q \rangle \\
e &::= v \mid \textstyle\sum_i \alpha_i e_i \mid \mathsf{let}\ p = \omega\, q\ \mathsf{in}\ e \\
\omega &::= f \mid \{v_i \leftrightarrow e_i\}_{i \in I} \mid \omega\,;\,\omega \mid \omega \oplus \omega \mid \omega \otimes \omega \mid \omega^{\dagger} \mid \mathsf{trace}_c\ \omega \\
t &::= e \mid \omega\, t \\
\mathit{decl} &::= f :: a \leftrightarrow b\,;\ f = \omega \mid \mathsf{round}\ f\ :\ a + c = \{v_i \leftrightarrow e_i\}_{i \in I}
\end{aligned}
$$

Fig. 13. Finite iso definitions. A name f refers to an earlier closed definition. The coefficient α_i is a complex scalar. A signature uses `::`, while the `round` abbreviation uses `:`. In trace_c, the return has type c.

The surface patterns exit v and back v elaborate recursively to left v and right v. This rewrite preserves payloads and nesting, and is performed before matching or typing by the core rules below. The remaining executable abbreviations are those of Section 2: `Unit`, `unit`, `Qubit` and its basis values, bracketed scalar combinations, combinator composition and `on` application. Listings typeset the source token `<->` as $\leftrightarrow$ and pair delimiters `<, >` as $\langle, \rangle$.

A.2 Statics

We check declarations in order. When a clause calls a named iso, its definition is already available. When it calls an iso written inside the clause, that iso is a smaller subprogram. We can therefore evaluate the call while checking the enclosing table, by induction on the declarations and their subprograms. Write $\Sigma\ \mathsf{ok}$ when all declarations in Σ have passed this check. A new name f must be fresh, and its definition must have its declared type using only the earlier bindings.

$$
\frac{}{\emptyset\ \mathsf{ok}} \qquad \frac{\Sigma\ \mathsf{ok} \quad f \notin \mathrm{dom}(\Sigma) \quad \Sigma \vdash \omega : a \leftrightarrow b}{\Sigma, (f = \omega : a \leftrightarrow b)\ \mathsf{ok}}.
$$

A `round` declaration checks the table at the same type on both sides and binds its name to the open body. Writing $\Sigma \vdash d \Rightarrow \Sigma'$ for declaration elaboration, we have

$$
\frac{\Sigma\ \mathsf{ok} \quad f \notin \mathrm{dom}(\Sigma) \quad \Sigma \vdash \mathscr{T} : (a + c) \leftrightarrow (a + c)}{\Sigma \vdash \mathtt{round}\ f : a + c = \mathscr{T} \Rightarrow \Sigma, (f = \mathscr{T} : (a + c) \leftrightarrow (a + c))}.
$$

An ordinary iso signature still permits distinct external types. The binding $f = \omega : a \leftrightarrow b$ represents the surface declaration $f :: a \leftrightarrow b\,;\ f = \omega$; listings may separate these parts by a newline. Write Δ for a finite set of distinct typed value variables. In Δ_1, Δ_2 the two domains must be disjoint. Each judgment below uses this convention, including inside a clause with an empty set of variables. The empty table presents the unique map on 0. Whenever we use a unit input or a one-dimensional span of exit maps, we assume the corresponding external type is inhabited.

Patterns and expressions. The finite basis is defined recursively:

$$\mathscr{B}(0) = \emptyset, \qquad \mathscr{B}(1) = \{*\},$$
$$\mathscr{B}(a+b) = \mathrm{left}\,\mathscr{B}(a) \sqcup \mathrm{right}\,\mathscr{B}(b), \quad \mathscr{B}(a \otimes b) = \mathscr{B}(a) \times \mathscr{B}(b).$$

An environment for Δ assigns each variable a basis value of its type. A variable matches any such value; a constructor matches its own tag, and a pair matches its two components. Write $\Delta \vdash v : a$ when a basis pattern is well typed. The rules are

$$\emptyset \vdash * : 1 \qquad x : a \vdash x : a \qquad \frac{\Delta \vdash v : a}{\Delta \vdash \mathrm{left}\ v : a+b} \qquad \frac{\Delta \vdash v : b}{\Delta \vdash \mathrm{right}\ v : a+b}$$

$$\frac{\Delta_1 \vdash v : a \qquad \Delta_2 \vdash w : b}{\Delta_1, \Delta_2 \vdash \langle v, w \rangle : a \otimes b}.$$

The basis assignments for a context Δ form $\mathscr{B}(\Delta) = \prod_{(x:a)\in\Delta} \mathscr{B}(a)$. The empty product contains the unique empty assignment. If any variable has an empty basis, the product is empty. Write $\mathscr{H}(\Delta) = \mathbb{C}^{\mathscr{B}(\Delta)}$. A product pattern p with $\Delta \vdash p : b$ gives a bijection $\gamma \mapsto p[\gamma]$ from $\mathscr{B}(\Delta)$ to $\mathscr{B}(b)$. The unit pattern sends the unique empty assignment to $*$, and a variable acts as the identity. For a pair we can recover the assignment by matching each component and joining the two assignments, since their contexts are disjoint. This gives the inverse by induction on p, including empty bases. A fixed tag preserves injectivity, so each match still gives a unique assignment.

The unit, variable and pair rules also type the product patterns p, q in Figure 13. In the argument position q these rules assemble already bound values. In the result position p they bind fresh variables of the indicated types. A single variable in either position may stand for a value of any type. For expressions we write $\Sigma; \Delta \vdash e : a$ and use

$$\frac{\Delta \vdash v : a}{\Sigma; \Delta \vdash v : a} \qquad \frac{\Sigma; \Delta \vdash e_i : a\ (i \in I) \qquad \alpha_i \in \mathbb{C}}{\Sigma; \Delta \vdash \sum_{i\in I} \alpha_i e_i : a}$$

$$\frac{\Delta_1 \vdash q : a \qquad \Sigma \vdash \omega : a \leftrightarrow b \qquad \Delta_p \vdash p : b \qquad \Sigma; \Delta_2, \Delta_p \vdash e : d}{\Sigma; \Delta_1, \Delta_2 \vdash \mathsf{let}\ p = \omega\, q\ \mathsf{in}\ e : d}.$$

In the last rule the variables in p are fresh and pairwise distinct. A product pattern of type b binds every basis output, by the bijection above. The sum rule permits an empty index set in any stated context Δ; its denotation is the zero linear map from $\mathscr{H}(\Delta)$. Such an expression cannot supply an entire column of a nonempty unitary table. A finite first-order term is a closed expression, or an application of a closed iso to a term of its domain. Writing $\Sigma \vdash t : a$ for a closed term, its rules are

$$\frac{\Sigma; \emptyset \vdash e : a}{\Sigma \vdash e : a} \qquad \frac{\Sigma \vdash t : a \qquad \Sigma \vdash \omega : a \leftrightarrow b}{\Sigma \vdash \omega\, t : b}.$$

This suffices for feeding a superposition of ground values to a program.

To check whether the table is reversible, we evaluate the clause that matches each basis input s and use its output as column s of M. Row s of $M^\dagger$ is the conjugate of that column, so

$$(M^\dagger M)_{ss'} = \sum_{t\in\mathscr{B}(b)} \overline{M_{ts}} M_{ts'}.$$

The diagonal entries give the squared column norms and the off-diagonal entries give the inner products between distinct columns. We can therefore check orthogonality on the expanded outputs.

Forming this matrix for a dense d-by-d table takes $O(d^3)$ arithmetic operations, where $d = 2^q$ for a register of q qubits.

$$\frac{\Delta_i \vdash v_i : a \quad \Sigma;\Delta_i \vdash e_i : b \quad (i \in I) \qquad \big(v_i[\mathscr{B}(\Delta_i)]\big)_{i\in I} \text{ disjointly covers } \mathscr{B}(a) \qquad |\mathscr{B}(a)| = |\mathscr{B}(b)| \quad M^\dagger M = I}{\Sigma \vdash \{v_i \leftrightarrow e_i\}_{i\in I} : a \leftrightarrow b}$$

Here $v_i[\mathscr{B}(\Delta_i)]$ is the set of all ground instances of pattern v_i. The coverage premise requires these sets to be disjoint and their union to be $\mathscr{B}(a)$. An empty instance set is permitted and contributes no column. This rule is exact over a specified coefficient domain with exact equality and arithmetic, for example $\mathbb{Q}(i, \sqrt{2})$, which contains the Hadamard coefficients. With mathematical complex parameters it is a well-formedness predicate; exact equality of arbitrary descriptions of complex numbers is not assumed decidable.

Reversible combinators. For $M \in \mathbb{C}^{\mathscr{B}(b)\times\mathscr{B}(a)}$ the atom rule is

$$\frac{|\mathscr{B}(a)| = |\mathscr{B}(b)| \qquad M^\dagger M = I}{[M] : a \leftrightarrow b}.$$

Write u, v for reversible programs and omit the common Σ in the following rules. A bound name has the type at its declaration. For finite types we have the structural isos of Π as abbreviations for the corresponding permutation clauses, including identity, symmetries and distributivity [30, § 2]. The reversible rules are

$$\frac{f : a \leftrightarrow b \in \Sigma}{f : a \leftrightarrow b} \qquad \frac{u : a \leftrightarrow b \qquad v : b \leftrightarrow d}{u\,;\,v : a \leftrightarrow d} \qquad \frac{u : a \leftrightarrow b}{u^\dagger : b \leftrightarrow a}$$

$$\frac{u : a \leftrightarrow b \qquad v : c \leftrightarrow d}{u \oplus v : a + c \leftrightarrow b + d} \qquad \frac{u : a \leftrightarrow b \qquad v : c \leftrightarrow d}{u \otimes v : a \otimes c \leftrightarrow b \otimes d} \qquad \frac{\beta : a + c \leftrightarrow b + c}{\mathrm{trace}_c\, \beta : a \leftrightarrow b}.$$

A.3 Dynamics

Evaluating an expression. Let $\gamma \in \mathscr{B}(\Delta)$ assign basis values to the variables in a linear context Δ, and write $\gamma \vdash e \Downarrow \psi$ when evaluation returns the finite output vector ψ. For a value pattern v, $v[\gamma]$ is its ground instance. Values and sums evaluate by

$$\frac{}{\gamma \vdash v \Downarrow |v[\gamma]\rangle} \qquad \frac{\gamma \vdash e_i \Downarrow \psi_i \ (i \in I)}{\gamma \vdash \sum_i \alpha_i e_i \Downarrow \sum_i \alpha_i \psi_i}.$$

For let $p = \omega q$ in e, split the input assignment as $\gamma_1 \uplus \gamma_2$ according to the typing rule: q uses precisely γ_1, and e retains γ_2. Write $U|q[\gamma_1]\rangle = \sum_t a_t|t\rangle$, where $\omega \Downarrow U$. Matching the product pattern p with t gives the fresh assignment γ_t. The application rule is

$$\frac{\omega \Downarrow U \quad \gamma_2 \uplus \gamma_t \vdash e \Downarrow \psi_t \quad (t \in \mathscr{B}(b))}{\gamma_1 \uplus \gamma_2 \vdash \text{let } p = \omega q \text{ in } e \Downarrow \sum_t a_t \psi_t}.$$

Here b is the output type of ω. Each alternative of a sum uses the same assignment. In a local call, only the bindings in γ_2 pass directly to the continuation. For closed terms, $t \Downarrow \psi$ uses the empty variable assignment, with

$$\frac{\varnothing \vdash e \Downarrow \psi}{e \Downarrow \psi} \qquad \frac{\omega \Downarrow U \qquad t \Downarrow \psi}{\omega t \Downarrow U\psi}.$$

The iso environment is fixed throughout. Names refer only to earlier definitions and Proposition 3.2 evaluates a trace in finitely many steps, even when infinitely many exits contribute. Write $E_e(s;\gamma)$ for the amplitude of output s when we evaluate e in environment γ. In a local application this gives

$$E_{\mathsf{let}}(s;\gamma_1,\gamma_2) = \sum_{t\in\mathscr{B}(b)} U_{t,q[\gamma_1]}\, E_e(s;\gamma_2,\gamma_t).$$

Here t ranges over intermediate basis outputs. Its amplitude is the entry of U in the column selected by $q[\gamma_1]$. Matching p to t gives exactly one assignment γ_t for the continuation. Multiplying its column by that amplitude and adding over t is ordinary matrix multiplication, with γ_2 carried through. There is no measurement or normalization between the two maps.

Elaboration and matrix evaluation. The matrix programs have grammar

$$\kappa ::= [M] \mid \kappa;\kappa \mid \kappa \oplus \kappa \mid \kappa \otimes \kappa \mid \kappa^\dagger \mid \mathsf{trace}_c\, \kappa.$$

The combinators κ of Section 2 have ground checked tables as atoms. Write $\mathscr{E}_\Sigma(\omega)$ for elaboration. If $f = \omega_f$ is a declaration in Σ, let Σ_f be its earlier prefix, the environment in which ω_f was checked. With M the evaluated columns of a table $\mathscr{T}$, the definition is

$$\begin{aligned}
\mathscr{E}_\Sigma(\mathscr{T}) &= [M], & \mathscr{E}_\Sigma(f) &= \mathscr{E}_{\Sigma_f}(\omega_f),\\
\mathscr{E}_\Sigma(\omega;\nu) &= \mathscr{E}_\Sigma(\omega);\mathscr{E}_\Sigma(\nu), & \mathscr{E}_\Sigma(\omega\oplus\nu) &= \mathscr{E}_\Sigma(\omega)\oplus\mathscr{E}_\Sigma(\nu),\\
\mathscr{E}_\Sigma(\omega\otimes\nu) &= \mathscr{E}_\Sigma(\omega)\otimes\mathscr{E}_\Sigma(\nu), & \mathscr{E}_\Sigma(\omega^\dagger) &= \mathscr{E}_\Sigma(\omega)^\dagger,\\
\mathscr{E}_\Sigma(\mathsf{trace}_c\,\omega) &= \mathsf{trace}_c\,\mathscr{E}_\Sigma(\omega).
\end{aligned}$$

In particular, elaborating `guardRound` gives the atom $[R_\theta]$. Elaborating `trace guardRound` gives $\mathsf{trace}_1[R_\theta]$. All these equations preserve types. Only `trace` connects the return wire.

If $\kappa \Downarrow M$ and $\lambda \Downarrow N$, the ordinary matrix rules, used with their typed premises, are

$$\begin{gathered}
[M] \Downarrow M, \qquad (\kappa;\lambda) \Downarrow NM,\\
(\kappa\oplus\lambda) \Downarrow M\oplus N, \quad (\kappa\otimes\lambda) \Downarrow M\otimes N, \quad \kappa^\dagger \Downarrow M^\dagger.
\end{gathered}$$

The feedback rule completes them:

$$\frac{\kappa \Downarrow U \quad U = \left(\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right) \quad F : X \to Y \quad (I_Y - D)F = C}{\mathsf{trace}_c\,\kappa \Downarrow A + BF}.$$

A column of F gives the returning amplitude for one basis input. The system can have several solutions when $I_Y - D$ is singular. Theorem 3.1 gives a solution and shows that every choice gives the same external matrix $A + BF$. Proposition 3.2 computes that matrix by eliminating one return pattern at a time. Write $\Sigma \vdash \omega \Downarrow M$ when its elaboration evaluates to M. We apply this matrix to a closed term by the rule

$$\frac{\Sigma \vdash \omega \Downarrow M \qquad t \Downarrow \psi}{\omega t \Downarrow M\psi}.$$

The calculation for a local call gives the induction step showing that expansion preserves every column of a clause. Values supply basis vectors and sums extend the calculation by linearity. Each combinator then acts on the same matrices before and after elaboration, including trace by Theorem 3.1. Equality on basis inputs extends linearly to inputs entangled with a retained factor.

A.4 Denotation

A checked table denotes the linear extension of its columns. Names, sequence, dagger, direct sum and tensor denote lookup, composition in execution order, adjoint, direct sum and tensor. The trace sums the exit maps in (3.2). The evaluation identities above give these same maps, with the trace case supplied by Theorem 3.1.

Control of an endo-iso u is $I \oplus u$ under the distributive iso for $(1+1)\otimes a$. Replacing u by $e^{i\varphi}u$ leaves its channel unchanged but gives just the second branch a phase. The channel alone therefore does not determine the controlled gate.

A.5 Effect extensions

Syntax. Write u for a reversible iso, v for an isometry program and h for a channel program. The two upper grammars are

$$v ::= \mathsf{arr}\, u \mid \mathsf{lift}\, u \mid v\,;\, v \mid v \oplus v \mid v \otimes v \mid \mathsf{count}_K\, u,$$
$$h ::= \mathsf{arr}\, v \mid \mathsf{hide}\, v \mid h\,;\, h \mid h \otimes h \mid \mathsf{while}_c\, u.$$

Statics. The type of the intermediate result must match the type expected by the next program. At the isometry layer this gives the following rules for sequence, tensor and direct sum.

$$\frac{v : a \rightarrowtail b \qquad w : b \rightarrowtail d}{v\,;w : a \rightarrowtail d} \qquad \frac{v : a \rightarrowtail b \qquad w : c \rightarrowtail d}{v \otimes w : a \otimes c \rightarrowtail b \otimes d} \qquad \frac{v : a \rightarrowtail b \qquad w : c \rightarrowtail d}{v \oplus w : a + c \rightarrowtail b + d}.$$

Channels also compose in sequence and by tensor product.

$$\frac{h : a \rightsquigarrow b \qquad k : b \rightsquigarrow d}{h\,;k : a \rightsquigarrow d} \qquad \frac{h : a \rightsquigarrow b \qquad k : c \rightsquigarrow d}{h \otimes k : a \otimes c \rightsquigarrow b \otimes d}.$$

These operations have their usual linear meanings, with sequence written in execution order. An identity on each layer is the embedded identity iso. The remaining rules give the types of allocation, hiding, the embeddings and the two loop constructs.

$$\frac{u : a \leftrightarrow b}{\mathsf{arr}\, u : a \rightarrowtail b} \qquad \frac{u : a + d \leftrightarrow b}{\mathsf{lift}\, u : a \rightarrowtail b} \qquad \frac{\beta : a + c \leftrightarrow b + c \qquad K \geq 1}{\mathsf{count}_K\, \beta : a \rightarrowtail [K] \otimes b}$$

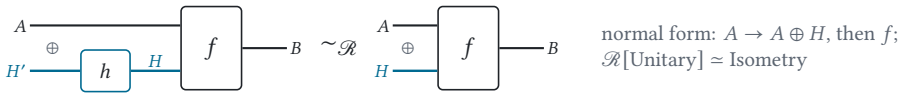
$$\frac{v : a \rightarrowtail b}{\mathsf{arr}\, v : a \rightsquigarrow b} \qquad \frac{v : a \rightarrowtail b \otimes g \qquad |\mathscr{B}(g)| > 0}{\mathsf{hide}\, v : a \rightsquigarrow b} \qquad \frac{\beta : a + c \leftrightarrow b + c}{\mathsf{while}_c\, \beta : a \rightsquigarrow b}.$$

In a mixed sequence we read an iso at the isometry layer as $\mathsf{arr}\,\omega$ and at the channel layer as $\mathsf{arr}(\mathsf{arr}\,\omega)$. The premise for while requires a reversible body. An inner coherent loop can be compiled first. A channel that measures its input, including another while, cannot occupy that position. At the top level both kinds of program accept a closed input term $t : a$. We distinguish their outputs because an isometry returns a vector while a channel returns an operator:

$$\frac{\Sigma \vdash t : a \qquad v : a \rightarrowtail b}{v\ \mathsf{on}\ t : \mathrm{Vec}(b)} \qquad \frac{\Sigma \vdash t : a \qquad h : a \rightsquigarrow b}{h\ \mathsf{on}\ t : \mathrm{Op}(b)}.$$

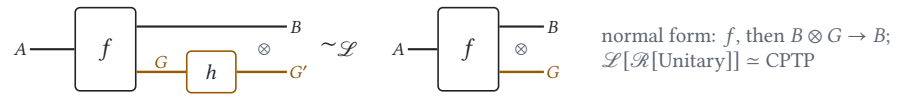
Denotation. Here $\mathrm{Vec}(b)$ consists of the vectors in $[\![b]\!]$ and $\mathrm{Op}(b)$ of the operators on that space. For $t \Downarrow \psi$, the two applications are $v\ \mathsf{on}\ t \Downarrow [\![v]\!]\,\psi$ and $h\ \mathsf{on}\ t \Downarrow [\![h]\!](|\psi\rangle\langle\psi|)$. For a unit vector the latter is a density matrix. The executable checks the input norm to numerical tolerance. An $\mathscr{R}[\mathsf{Unitary}]$ representative $U : X \oplus H \to X'$ exposes $u \mapsto U(u, 0)$, identifying changes confined to the unused summand. An $\mathscr{L}[\mathsf{Isometry}]$ representative $V : X \to X' \otimes G$ exposes $\rho \mapsto \mathrm{tr}_G(V\rho V^\dagger)$, identifying isometric relabelings of the hidden factor and the equivalence relation they generate [27, Eqs. (3)–(4), Thm. 22].

(a) $\mathscr{R}[\mathscr{C}]$: a heap enters beside the input, and only access to it counts



normal form: $A \to A \oplus H$, then f;
$\mathscr{R}[\text{Unitary}] \simeq$ Isometry

(b) $\mathscr{L}[\mathscr{C}]$: garbage leaves beside the output, and its contents do not count



normal form: f, then $B \otimes G \to B$;
$\mathscr{L}[\mathscr{R}[\text{Unitary}]] \simeq$ CPTP

Fig. 14. Allocation exposes an input summand of a larger reversible map. Hiding exposes one output factor of an isometry. Preprocessing the unused summand or isometrically relabeling the hidden factor preserves the exposed map. Here $\mathscr{C} = \text{Unitary}$ in $\mathscr{R}[\mathscr{C}]$ and $\mathscr{C} = \text{Isometry}$ in $\mathscr{L}[\mathscr{C}]$.

To allocate the basis value $j < K$, choose the permutation $p_j : a + ([K-1] \otimes a) \leftrightarrow [K] \otimes a$ that sends left x to $\langle j, x\rangle$, with $0 \le j$ and $[0] = 0$. Then $\text{alloc}_j = \text{lift}\, p_j$. For the basis copier on inhabited b, the heap $[|\mathscr{B}(b)| - 1] \otimes b$ supplies the $|\mathscr{B}(b)|^2 - |\mathscr{B}(b)|$ inputs needed to complete the diagonal injection to a permutation.

An embedded unitary acts by the same map. Embedding an isometry V gives the channel $\mathscr{E}_V$. We reserve $\mathscr{E}_\Sigma$ for elaboration of syntax. To form count_K, apply $I_{[K]} \otimes \beta$ and distribute the counter over the two tags. Apply S_K on the returning tag alone. The blocks are now $I \otimes A, I \otimes B, S_K \otimes C, S_K \otimes D$, so exit k is $S_K^k \otimes x_k$. Closing $[K] \otimes c$ and preparing the counter at zero gives (5.1).

Since the counter comes first, hiding it gives $\text{hide}(\text{count}_K\, \beta; \text{arr}\, \sigma_{[K],b})$. In executable syntax `count K beta` returns data first, so `discard` already hides its counter and `swap` restores the displayed order. Here `discard` is the channel $I_a \otimes \text{discard}_b$; $\text{unit}_\otimes$ built from whole-register discard $\text{discard}_b : b \rightsquigarrow 1$.

For the empty type, discard_0 is obtained by hiding the unit factor of the unique isometry $0 \rightarrowtail 1 \otimes 1$, constructed by lift from $0 + 1 \leftrightarrow 1 \otimes 1$. Its action is the unique map on the zero operator space. Set measure_0 to the embedded identity on 0. These supply the empty cases omitted from the inhabited-type formulas in the body.

For while, allocate a fresh bit at 0 and apply the ordinary iso

$$\omega_{\text{tag}} = \{ \begin{array}{l} \langle \text{exit}\, x, 0\rangle \leftrightarrow \langle \text{exit}\, x, 0\rangle, \\ \langle \text{exit}\, x, 1\rangle \leftrightarrow \langle \text{exit}\, x, 1\rangle, \\ \langle \text{back}\, y, 0\rangle \leftrightarrow \langle \text{back}\, y, 1\rangle, \\ \langle \text{back}\, y, 1\rangle \leftrightarrow \langle \text{back}\, y, 0\rangle \quad \} \end{array} : (b + c) \otimes \texttt{Qubit} \leftrightarrow (b + c) \otimes \texttt{Qubit}.$$

The clauses permute the basis. Fixing the fresh bit at 0 gives the tag isometry T. On $v \in [\![b]\!]$ and $y \in [\![c]\!]$ it acts by

$$T(v, y) = (\iota_{\text{ext}} v) \otimes |0\rangle + (\iota_{\text{ret}} y) \otimes |1\rangle.$$

Applying T after U and tracing out the copied bit gives the diagonal output blocks. Feedback now uses the classical sum. This construction works when the payload types differ. The notation while names this classical feedback operation on the translated body.

When $a = b = c$, let $\text{factor} : a + a \leftrightarrow a \otimes \texttt{Qubit}$ and $\text{dist} = \text{factor}^\dagger$, and put $\hat{\beta} = \text{dist}; \beta; \text{factor}$. Writing $E(\omega) = \text{arr}(\text{arr}\, \omega)$, the channel in Figure 10, written here with the payload first, is

$$E(\text{factor}); E(\hat{\beta}); (I_a \otimes \text{measure}_{\texttt{Qubit}}); E(\text{dist}).$$

The first factor; dist cancel, leaving the original body followed by exposure, measurement and restoration of its tag.

Soundness. We check definitions in order. For each basis input we evaluate the matching clause under its bindings to obtain one column. The table denotes a unitary because the admission rule requires these columns to form an orthonormal basis of the output. Sequence, sum, tensor and dagger preserve unitarity, as does trace by Theorem 3.1.

For lift, sending u to $(u, 0)$ preserves inner products, as does the unitary that follows. The same argument applies to count_K, which sends u to $|0\rangle \otimes u$ before applying the enlarged loop. Embedding an isometry gives a channel, and tracing out one output preserves that property. For `while`, each partial sum in (5.2) is completely positive because each exit acts by $\rho \mapsto x_k \rho x_k^\dagger$. The cone is closed in finite dimension, so the limit is completely positive too. The limit preserves trace because the exit probabilities sum to one on every unit input, by $\sum_k x_k^\dagger x_k = I$.

Consequently, induction on the checked program gives a unitary for an iso, an isometry for an allocation program and a channel for a hiding program. In each case evaluation computes its denotation: the clause case agrees on every basis column, the structural rules apply the corresponding matrix operations, and the loop cases use their exit maps. A closed term $t : a$ denotes a vector in $\mathbb{C}^{\mathscr{B}(a)}$ whose norm depends on its coefficients. Conversely, given a finite unitary V, we can write its columns as the clauses $s \leftrightarrow \sum_t V_{ts} t$. They pass the admission check since they form an orthonormal basis, also when both basis sets are empty.

B Convergence, probability and elimination

A permutation gives each value a unique predecessor, so a path from outside cannot enter an internal cycle. Its first value on the cycle would already have a predecessor there. The path therefore exits within $|\mathscr{B}(c)| + 1$ calls. The two external paths in Figure 15 combine to a swap, as in reversible flowchart semantics [20, Prop. 3.10][32, Prop. 5.3]. A unitary body can send some amplitude back on every call. To sum its exits, we separate the internal states that can still emit from those that stay inside forever.

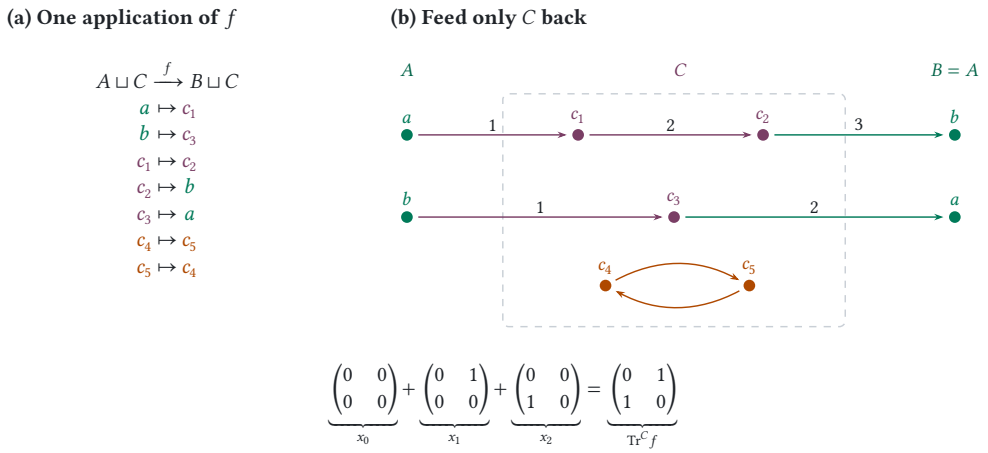


Fig. 15. A reversible loop with two external paths and an internal cycle. An edge numbered $k + 1$ is the call after k returns. The two nonzero exit maps have disjoint domains and their sum is a swap.

The split illustrated by `extraPort` in Section 3 lets us bound the weight left after each call.

Proof of Theorem 3.1. If we start with y on the return wire, the next call emits By and sends Dy back. After j more returns the amplitude that can leave is $BD^j y$, so the states that never emit form $Y_0 = \bigcap_{j\ge 0} \ker BD^j$, the unitary part of D in the Sz.-Nagy–Foiaş decomposition [3, Thm. 3.3.7].

For $y \in Y_0$ the whole output of U is $(0, Dy)$, with $Dy \in Y_0$. Preservation of norm makes D an isometry on Y_0, hence onto there by finite dimensionality. Since U carries $0 \oplus Y_0$ onto itself, its restriction there is a unitary D_0 whose eigenvectors span Y_0. Conversely, if $Dv = \lambda v$ with $|\lambda| = 1$, norm preservation gives $Bv = 0$, hence $BD^j v = 0$ for every j. This identifies the two descriptions of Y_0. On the orthogonal complement, U restricts to the unitary U_1 that takes $X \oplus Y_1$ onto $X' \oplus Y_1$, where $Y_1 = Y_0^{\perp}$. In particular, the first return Cu lies in Y_1 and D keeps every later return there.

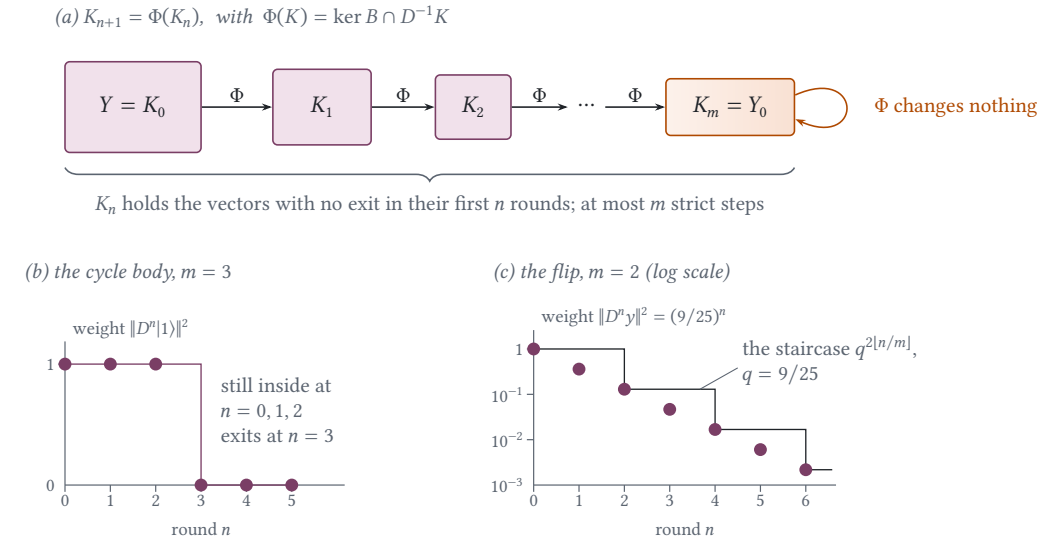


Fig. 16. The no-emission spaces stabilize within $\dim Y$ steps. Panel (b) is the four-state cycle $0 \mapsto 1 \mapsto 2 \mapsto 3 \mapsto 0$, with external state 0. Its three-dimensional return block has $D|1\rangle = |2\rangle$, $D|2\rangle = |3\rangle$ and $D|3\rangle = 0$. In panel (c), $\|D_1\| = 3/5$ and $m = 2$, so $q = \|D_1^2\| = 9/25$. Each pair of calls multiplies the weight by q^2, so the staircase $q^{2\lfloor n/m \rfloor}$ meets the exact weight $(9/25)^n$ at every even n.

To see how quickly amplitude leaves Y_1, let K_j consist of the states that emit nothing for the first j calls. Then $K_0 = Y$ and $K_{j+1} = \{y : By = 0,\ Dy \in K_j\}$. Each space lies inside the one before it. If two successive spaces agree, the recurrence gives the same space at every later call. There can be at most $m = \dim Y$ strict decreases, so $K_m = Y_0$ (Figure 16).

Write $D_1 = D|_{Y_1}$ and $B_1 = B|_{Y_1}$. If $Y_1 = 0$, then $C = 0$, every delayed exit vanishes and we can take $F = 0$. Otherwise $m > 0$. A unit vector $y \in Y_1$ must emit on at least one of its first m calls, since it lies outside K_m. Conservation of norm gives

$$1 - \|D_1^m y\|^2 = \sum_{j=0}^{m-1} \|BD^j y\|^2 > 0.$$

The unit sphere is compact, so even the largest remaining norm $q = \|D_1^m\|$ is less than one. After $k = pm + r$ calls, with $0 \le r < m$, the first p groups of m calls leave norm at most q^p, and the last r calls cannot increase it. This bounds $\|D_1^k\|$ by q^p.

By this bound the sum $F = \sum_{j\ge 0} D_1^j C$ converges. Its first term is C and the remaining terms sum to $D_1 F$, so $F = C + D_1 F$. Applying B_1 sends each return to its exit. These delayed exits are absolutely

summable even with a polynomial weight in j. Adding the immediate exit gives $L = A + BF$. If two solutions differ by w, subtracting their wire equations gives $Dw = w$. On input $(0, w)$ the body therefore outputs (Bw, w). Since the returning component has the full input norm, $Bw = 0$ and the two solutions give the same external output.

For the solution Fu, the body sends (u, Fu) to (Lu, Fu). Canceling $\|Fu\|^2$ from the two squared norms gives $\|Lu\| = \|u\|$. The external spaces have the same finite dimension because U is unitary, so L is onto as well.

To account for the probability, let $T_j = C^\dagger D^{\dagger j} D^j C$. On a unit input u, $\langle u, T_j u\rangle$ is the probability still inside after call $j+1$. The first call gives $x_0^\dagger x_0 = I - T_0$. On later calls the weight that leaves is exactly the drop in the weight still inside, since

$$x_{j+1}^\dagger x_{j+1} = C^\dagger D^{\dagger j} \underbrace{B^\dagger B}_{I - D^\dagger D} D^j C = T_j - T_{j+1}.$$

Adding through exit k gives $\sum_{j=0}^{k} x_j^\dagger x_j = I_X - T_k$. The residual term tends to zero by the decay just proved, giving total exit probability one even for a body with infinitely many possible exits. □

A state can stay inside forever without being fixed. For example, if $Dv = -v$ then every call preserves its norm, so $Bv = 0$. Only a nonzero state with $Dv = v$ makes $I - D$ singular at phase one.

B.1 Eliminating a pattern

Closing one port requires summing its repeated returns. When its self-entry differs from one, this gives the usual scattering-matrix update [22, § 4, Thm. 12] [46, App. A, Eq. (A1)]. When the self-entry equals one, unitarity tells us what to do instead.

Proof of Proposition 3.2. Choose the returning basis value q we want to remove, and leave the other return wires open for this step. Write $Q = \mathbb{C}q$ and $Y_{\text{rest}} = Y \ominus Q$. The ports left open form $E_{\text{in}} = X \oplus Y_{\text{rest}}$ and $E_{\text{out}} = X' \oplus Y_{\text{rest}}$. In this order the body has matrix

$$\begin{pmatrix} M & b_q \\ c_q & d_q \end{pmatrix}, \qquad M : E_{\text{in}} \to E_{\text{out}}, \quad b_q : Q \to E_{\text{out}}, \quad c_q : E_{\text{in}} \to Q, \quad d_q = D_{qq}.$$

On input $w = (u, y_{\text{rest}})$, the other ports send amplitude $c_q w = (Cu)_q + \sum_{r \neq q} D_{qr} y_r$ to q. If λ comes back on q itself, the same port receives a further $d_q\lambda$. Closing this one wire therefore asks us to solve $\lambda = c_q w + d_q \lambda$. The other ports receive $Mw + b_q\lambda$.

If $d_q \neq 1$, substitution gives the new table $M + b_q c_q/(1 - d_q)$, the Schur complement of $d_q - 1$ in the matrix with blocks $M, b_q, c_q, d_q - 1$. This division also works for a phase such as -1 or i. When $d_q = 1$, that entry accounts for the whole squared norm of its row and column, so $c_q = b_q = 0$. The wire equation reduces to $\lambda = \lambda$, and the other ports see Mw for every choice of λ. In either case the new table is $\mathrm{Tr}^Q(U)$, which is unitary by Theorem 3.1.

For `guardRound` there is only one returning value. Substituting $M = d_q = 3/5$, $b_q = -4/5$ and $c_q = 4/5$ gives the single coefficient $3/5 - (4/5)^2/(1 - 3/5) = -1$ of the clause in Section 3.3. In `flipRound`, the updated table still contains `back 1`; its new self-entry $9/25$ is the d_q for the next step (Figure 5).

A solution on all the wires restricts to a solution on the remaining wires by the substitution just made. Conversely, from a solution on those wires we recover λ by the same division when $d_q \neq 1$ and may choose $\lambda = 0$ when $d_q = 1$. Thus each step preserves the external output. Each step also removes one basis value and uses only field operations with a nonzero divisor. In any order there are $|\mathscr{B}(c)|$ steps, ending with a table over the original field for the gate L. The same argument applies after putting a phase on the return wire, since the modified body is again unitary. □

Closed sums for the running rotation. For `guardRound`, both port spaces are $\mathbb{C}$, so the blocks are scalars: $A = D = t = 3/5$, $C = s = 4/5$ and $B = -s$. The general exit formula becomes $x_0 = t$ and $x_{k+1} = (-s)t^k s = -s^2 t^k$. To compare adding these amplitudes with adding their squared magnitudes, follow the first n calls on unit input. Write r_n for the amplitude still returning, ℓ_n for the sum emitted so far, and p_n for the weight of those exits when the guard is measured. The geometric sums give

$$r_n = st^{n-1},$$
$$\ell_n = t - s^2 \sum_{j=0}^{n-2} t^j = -1 + (1+t)t^{n-1},$$
$$p_n = t^2 + s^4 \sum_{j=0}^{n-2} t^{2j} = 1 - s^2 t^{2(n-1)}.$$

The sums are empty when $n = 1$. Table 2 follows the resulting exits, residuals and sums. The squared length $|\ell_n|^2$ can fall and rise as later exits interfere. The measured halting weight p_n only increases, because each call adds a nonnegative probability.

	one call		coherent feedback		guard measured
call n	emitted x_{n-1}	still returning r_n	sum emitted ℓ_n	squared length $\lvert\ell_n\rvert^2$	halted so far p_n
1	$\frac{3}{5}$	$\frac{4}{5}$	$\frac{3}{5}$	0.3600	0.3600
2	$-\frac{16}{25}$	$\frac{12}{25}$	$-\frac{1}{25}$	0.0016	0.7696
3	$-\frac{48}{125}$	$\frac{36}{125}$	$-\frac{53}{125}$	0.1798	0.9171
4	$-\frac{144}{625}$	$\frac{108}{625}$	$-\frac{409}{625}$	0.4282	0.9701
5	$-\frac{432}{3125}$	$\frac{324}{3125}$	$-\frac{2477}{3125}$	0.6283	0.9893
$n \to \infty$	0	0	-1	1	1

Table 2. `guardRound` on $|*\rangle$. Coherent feedback adds exits, while guard measurement adds their squared magnitudes. A record keeps exit x_{n-1} with label $n-1$. Only p_n is a halting probability; decimal entries are rounded to four places.

A table and an unknown gate. For a scalar rotation with cosine $-1 < t < 1$, the trace is $t - (1-t^2)/(1-t) = -1$. At the identity rotation it is $+1$. With a bypass and a final Hadamard, the probability of reading 1 therefore jumps from one to zero at the identity. In any fixed finite circuit built by composing the supplied gate, this probability varies continuously. If it gives probability p at the identity, its error there is p and its error on nearby rotations tends to $1 - p$. At least one of these is $1/2$ or more. Given the clause table, elimination can test $t = 1$ exactly [17, Remark B.1]. Section 4 follows the light from its first trip around an empty ring.

C Records, measurement and phase

In `count 2 guardRound`, a return flips the bit and an exit keeps its current value. For a counter of size K, the flip becomes the cyclic successor S_K on the returning output.

For general ports, distributivity d_K identifies $[K] \otimes (a + c)$ with $([K] \otimes a) + ([K] \otimes c)$. Write d'_K for the same distributivity at the output. The enlarged body of Figure 8 is

$$\beta_K = d_K^{-1} \,;\, (I_{[K]} \otimes \beta) \,;\, d'_K \,;\, (I_{[K]\otimes b} \oplus (S_K \otimes I_c)).$$

The upper row of its matrix sends amplitude out with the current label. The lower row applies S_K before sending amplitude back.

$$W_K = [\![\beta_K]\!] = \begin{pmatrix} I_K \otimes A & I_K \otimes B \\ S_K \otimes C & S_K \otimes D \end{pmatrix}. \qquad R_K u = (\mathrm{Tr}^{\mathbb{C}^K \otimes Y} W_K)(|0\rangle \otimes u).$$

When both ports carry the same payload $P = X = X' = Y$, the sheet diagram draws the guard and data on one sheet. The change of coordinates $F_g = \mathrm{factor}; \sigma_{P,\mathrm{Qubit}} : P + P \leftrightarrow \texttt{Qubit} \otimes P$ puts the guard first, giving $\widehat{U}_g = [\![F_g]\!] U [\![F_g]\!]^{-1}$. With a counter, $F_{K,g} = d_K^{-1}; (I_{[K]} \otimes F_g)$ puts counter, guard and data in that order. Its inverse restores the two ports.

On the first call the input is $(|0\rangle \otimes u, 0)$. The ket is the prepared counter and the second component 0 means no returning amplitude. The two output blocks are $|0\rangle \otimes Au$ and $|1 \bmod K\rangle \otimes Cu$.

Figure 17 puts the counter increment and the tag-copying step beside the same body.



Fig. 17. One call to the same body U, followed by an operation controlled on the outgoing guard $g = 1$. A filled dot denotes a coherent control. The left circuit carries j into the next round. In the right circuit the discarded bit distinguishes the tags without distinguishing states of the payload q. Crossing wires interact only at a marked gate.

From a finite counter to the exact count. Give each return count its own basis state in $H_{\mathrm{rec}} = \ell^2(\mathbb{N})$. With the counter first, $Ru = \sum_k |k\rangle \otimes x_k u$ has squared norm $\sum_k \|x_k u\|^2 = \|u\|^2$. For `guardRound` on $|*\rangle$ the data factor is always $|*\rangle$, and the count state is $\xi_{\mathrm{rot}} = t|0\rangle - s^2 \sum_{k \geq 1} t^{k-1}|k\rangle$, with $t = 3/5$ and $s = 4/5$. Section 6 factors out $L|*\rangle = -|*\rangle$, so its count vector is $-\xi_{\mathrm{rot}}$. In the bypass experiment we factor out $|1\rangle$ on the rotating path, so its count state is ξ_{rot}.

In `bypassRound`, the bypass exits at once, so $R|0\rangle = |0\rangle \otimes |0\rangle$, while $R|1\rangle = \xi_{\mathrm{rot}} \otimes |1\rangle$. Only the rotating arm has a nonzero $|1\rangle$ count amplitude, so $R|+\rangle$ is entangled. With a K-state counter, the rotating arm's amplitude at label zero is $g_0 = t - s^2 t^{K-1}/(1 - t^K)$, since all positive multiples of K reach that label. The bypass keeps $|0\rangle$, so g_0 is the overlap of the two unit count states. At output 1 the final Hadamard gives half their difference. Its squared norm is $(2 - 2g_0)/4 = (1 - g_0)/2$. For $t = 3/5$ this gives one at $K = 1$ and $1/2$ at $K = 2$. Since $g_0 < t$ and $g_0 \to t$, it approaches the measured value $(1 - t)/2 = 1/5$ from above.

A caller holding the counter can perform a quantum eraser measurement in the Fourier basis. Projecting onto $K^{-1/2} \sum_{j=0}^{K-1} |j\rangle$ gives $K^{-1/2} \sum_k x_k u = Lu/\sqrt{K}$. This outcome has probability $1/K$ on every unit input. When the body is known, reversing the enlarged loop recovers $|0\rangle \otimes u$. Applying L to the data then gives Lu with certainty (Section 5).

Embed the K counter labels as the first K basis states by $J_K : \mathbb{C}^K \to H_{\mathrm{rec}}$. The finite and exact records agree term by term until a label wraps. Their difference is bounded by the remaining exits.

$$\|(J_K \otimes I)R_K - R\| \leq \sum_{k \geq K} \big\| |k \bmod K\rangle - |k\rangle \big\| \, \|x_k\| \leq \sqrt{2} \sum_{k \geq K} \|x_k\|.$$

For $k \geq K$ the two labels are orthogonal unit vectors, at distance $\sqrt{2}$. Theorem 3.1 gives absolute convergence, so this tail tends to zero.

Folding counts uses the absolute summability of the exits. On an arbitrary record with amplitude $1/\sqrt{n}$ at each of $0, K, \dots, (n-1)K$, the norm is one. Folding modulo K adds all n terms at zero, giving $\sqrt{n}|0\rangle$. Summing at phase one also gives $\sqrt{n}$, so both operations are unbounded on ℓ^2.

The exact count can also be written as a sequence of guard outcomes. In Ying's free coin histories, k returns followed by an exit give the word $J|k\rangle = |1^k 0\rangle$ in the $(k+1)$-coin sector [52, Eq. (34)]. These words are orthonormal, so J identifies the exact register isometrically with their closed span. For matched exit maps, $(J \otimes I)Ru$ and Ru give the same data when we discard the record. This comparison concerns completed histories. Ying's principal-system semantics also specifies an initial coin state and uses the symmetric semantics [52, Def. 9]. Assolini and Di Pierro keep a qubit from each test. Their unitary approximations can fail to converge while a branch keeps running, although the output on completed branches converges for each input [4, § 4.1, Thm. 5.9].

A clock can also keep the count. One tick of delay per return puts $x_0 u, x_1 u, \dots$ in consecutive, orthogonal output slots [3, § 3.4]. At a common observation time, a shared sequence of post-halt ancilla states can likewise distinguish halt times. Branch-dependent sequences can let those histories interfere again [34]. For an exact record, the labels must still be orthogonal when observed. In the ring, a short pulse separates the exits in time, while steady light at one frequency brings them to the same mode (Appendix H).

What remains after hiding the count. For an input of the form $|u\rangle\langle v|$, tracing the exact record over its orthonormal basis leaves $\sum_k x_k |u\rangle\langle v| x_k^\dagger$. Such operators span the input matrix algebra, giving (5.2). For `bypassRound`, all delayed exits are multiples of $P = |1\rangle\langle 1|$. Their squared coefficients sum to $s^4/(1-t^2) = s^2$. Thus $u \mapsto |0\rangle \otimes Au + |1\rangle \otimes sPu$ also gives the measured channel when we hide the bit. This bit represents two Kraus operators. It carries neither the count nor its parity.

For a finite counter put $G_j = \sum_{k \equiv j \bmod K} x_k$. Discarding that counter instead gives $\sum_{j<K} G_j \rho G_j^\dagger$. As a result, unequal counts congruent modulo K retain cross terms. For the Fourier forms, the coefficient sum is absolutely convergent, and

$$G_j = \frac{1}{K} \sum_{z^K=1} \bar{z}^j \Theta(z), \qquad x_k = \frac{1}{2\pi} \int_0^{2\pi} e^{-ik\varphi} \Theta(e^{i\varphi})\, d\varphi.$$

At phase one the exits add to L. Averaging $\Theta(z)\rho\Theta(z)^\dagger$ over the circle cancels every term with unequal counts, giving (7.1) by absolute convergence. Over the Kth roots, a pair of exits k, j instead acquires the factor $K^{-1} \sum_{z^K=1} z^{k-j}$. It is one if $k \equiv j \pmod K$ and zero otherwise, so the average is $\sum_{r<K} G_r \rho G_r^\dagger$, the data channel of the hidden K-counter. If $D^n = 0$, only counts $0, \dots, n$ can occur, so $n+1$ equally spaced phases give the measured loop exactly.

We can recover the measured channel by closing the return block of a single measured call, as in Figure 11. Let $\theta(\rho, \sigma) = \mathrm{diag}(\rho, \sigma)$ insert the two operator blocks and let φ extract the diagonal output blocks. Then $\Phi_U = \varphi \circ \mathscr{C}_U \circ \theta$ is the branching map of Section 5 [3, Def. 3.3.18]. In CPTR, the category of finite-dimensional C^*-algebras and completely positive trace-nonincreasing maps, the classical-sum trace $\mathrm{Tr}_{\mathrm{cl}}^{\mathrm{End}(Y)}(\Phi_U)$ sends ρ to $\sum_k x_k \rho x_k^\dagger$ [43, § 5.1, Eq. (1)] [3, Thm. 3.2.11, Prop. 3.2.17].

It is the extraction of the diagonal blocks after each call that removes coherence between exit and return. Embedding the unitary in finite isometries and tracing there still gives the coherent loop [3, Cor. 3.3.16].

Relation to monitored first returns. The published test-first loop measures before applying U [53, § 3]. To obtain our body-first convention when $X' = X$, apply U once before entering it. Let

$\iota_X u = (u, 0)$ and $P = \iota_X \iota_X^\dagger$. For $n \geq 1$, the exit map on call n is $\iota_X^\dagger U((I-P)U)^{n-1}\iota_X = x_{n-1} : X \to X$. Thus the monitored-return count includes the initial call and is one more than our number of returns [9, § 2].

To see where the least fixed point enters, consider the state σ just before the test. The exit contributes $P\sigma P$. On a return, the next call prepares $V\sigma V^\dagger$ for another test, where $V = U(I-P)$. Thus Ying's unfolding sends a map G to $\sigma \mapsto P\sigma P + G(V\sigma V^\dagger)$ [51, arXiv v1, Prop. 5.1(6), Cor. 5.1].

Starting from zero, n unfoldings give the first n exits, $\sum_{j=0}^{n-1} PV^j\sigma(V^j)^\dagger P$. For positive σ each new exit adds a positive operator. Since $V^\dagger V = I - P$, their total trace is $\operatorname{tr}\sigma - \operatorname{tr}(V^n\sigma(V^n)^\dagger)$. The partial sums are thus increasing with bounded trace, so they converge in finite dimension. Passing to the limit in the recurrence gives a solution. It is the least one because any positive solution bounds the initial zero and, by order preservation at each unfolding, every partial sum. Starting with $U\iota_X\rho\iota_X^\dagger U^\dagger$ and reading the external block gives precisely $\sum_k x_k \rho x_k^\dagger = \mathscr{M}(\rho)$.

D Agreement and replacement

Proof of Theorem 6.1. Suppose the measured and coherent loops give the same channel. On a unit input u the coherent loop returns Lu, so a test for any unit vector w orthogonal to Lu has probability zero. The same test on the measured loop gives $\sum_k |\langle w, x_k u\rangle|^2$. For this sum to vanish, each term must vanish. Thus every $x_k u$ lies along Lu.

So far the multiplier could depend on the input. For a fixed exit k, put $V = L^\dagger x_k$, so that Vu lies along u for every u. In particular, $Ve_i = \lambda_i e_i$ on each basis vector. On $e_i + e_j$ this gives

$$V(e_i + e_j) = \lambda_i e_i + \lambda_j e_j = \lambda_{ij}(e_i + e_j).$$

Comparing the two coordinates gives $\lambda_i = \lambda_j$. Thus $V = b_k I$ and $x_k = b_k L$ for one scalar b_k on the whole input space. In dimension one there is only one coefficient to begin with.

What stays in the count register now follows from the two ways of summing exits. Since $\sum_k x_k = L \neq 0$, absolute convergence gives $\sum_k b_k = 1$. Since $\sum_k x_k^\dagger x_k = I$, we also have $\sum_k |b_k|^2 = 1$. The exact counter therefore returns a unit vector $\xi = \sum_k b_k|k\rangle$, the same for every input, and $Ru = \xi \otimes Lu$. Conversely, discarding such a factor gives $\mathscr{E}_L$. Taking the kth coordinate recovers $x_k = \langle k|\xi\rangle L$ for every exit.

It remains to decide this condition from finitely many exits. The immediate exit is $x_0 = A$. For $m = \dim Y > 0$, the next m exits are $BC, BDC, \ldots, BD^{m-1}C$. By Cayley–Hamilton, D^m is a linear combination of $I, D, \ldots, D^{m-1}$, and multiplying by D gives the same recurrence for all higher powers. Hence every later exit lies in the span of those m delayed exits. When $m = 0$, only the immediate exit can be nonzero.

The span of $x_0, \ldots, x_m$ thus contains every exit and, being finite-dimensional, also their sum L. On a nonzero input space L is nonzero. If this span has dimension at most one, it must be the line through L, so every x_k is a multiple of L. Conversely, multiples of L pass the finite test. □

For a dense table with $n = \dim X = \dim X'$ and $m = \dim Y$, start with $P = C$. Forming BP gives the next exit and replacing P by DP gives the next return. Forming the m delayed exits this way costs $O(m^2n^2 + m^3n)$ field operations. The entry comparisons in Section 6 then test their proportionality together with A.

We can also compute the measured channel directly. Restrict B, D to the emitting space Y_1 as in Appendix B, and write $\mathscr{E}_T(\rho) = T\rho T^\dagger$. On return density matrices the next call acts by $\mathscr{E}_{D_1}$. Its powers decay, so summing the delayed exits gives

$$\mathscr{M} = \mathscr{E}_A + \mathscr{E}_{B_1}(I - \mathscr{E}_{D_1})^{-1}\mathscr{E}_C.$$

Comparing this with $\mathscr{E}_L$ on matrix units is a second finite test. For two loops with separate exact counts, forgetting both records composes their measured channels. With one shared exact count, exits with the same total add before we forget it, so we apply the formula to the combined body of Section 8. Comparing these channels on matrix units decides whether sharing changes the data.

The bypass shows why we test proportionality exactly. As t approaches one from below, $x_1 = -(1-t^2)|1\rangle\langle 1|$ tends to zero. Yet the coherent caller still reads 1 with certainty, while measuring the guard gives 1 with probability $(1-t)/2$. As that probability tends to zero, rounding x_1 to a zero multiple of x_0 would accept a change the caller can almost certainly detect.

Channel contexts. Fix the hole type $a \rightsquigarrow b$ and abbreviate it by $\mathscr{H}$ to the left of a context judgment. For fixed closed channels h, h_j, the contexts are generated by

$$\begin{aligned}\mathscr{C} \;::=\; & [-] \mid h; \mathscr{C} \mid \mathscr{C}; h \mid h \otimes \mathscr{C} \mid \mathscr{C} \otimes h \\ & \mid \mathsf{case}\big(\Psi_i; \mathscr{C}, (\Psi_j; h_j)_{j\neq i}\big).\end{aligned}$$

The rules for the hole, sequence and tensor are

$$\mathscr{H} \vdash [-] : a \rightsquigarrow b \qquad \frac{h : e \rightsquigarrow c \qquad \mathscr{H} \vdash \mathscr{C} : c \rightsquigarrow d}{\mathscr{H} \vdash h; \mathscr{C} : e \rightsquigarrow d} \qquad \frac{\mathscr{H} \vdash \mathscr{C} : c \rightsquigarrow d \qquad h : d \rightsquigarrow e}{\mathscr{H} \vdash \mathscr{C}; h : c \rightsquigarrow e}$$

$$\frac{h : e \rightsquigarrow f \qquad \mathscr{H} \vdash \mathscr{C} : c \rightsquigarrow d}{\mathscr{H} \vdash h \otimes \mathscr{C} : e \otimes c \rightsquigarrow f \otimes d} \qquad \frac{\mathscr{H} \vdash \mathscr{C} : c \rightsquigarrow d \qquad h : e \rightsquigarrow f}{\mathscr{H} \vdash \mathscr{C} \otimes h : c \otimes e \rightsquigarrow d \otimes f}.$$

For a finite instrument $(\Psi_j : c \to c_j)_{j\in J}$, meaning $\Psi_j : \mathrm{End}([\![c]\!]) \to \mathrm{End}([\![c_j]\!])$, the maps are completely positive and their output traces sum to the input trace. Placing each output in its summand of $\bigoplus_j [\![c_j]\!]$ gives an instrument with a common output space. Its case rule is

$$\frac{i \in J \quad \mathscr{H} \vdash \mathscr{C} : c_i \rightsquigarrow d \quad h_j : c_j \rightsquigarrow d \; (j \neq i)}{\mathscr{H} \vdash \mathsf{case}\big(\Psi_i; \mathscr{C}, (\Psi_j; h_j)_{j\neq i}\big) : c \rightsquigarrow d}.$$

Plugging h into the hole interprets sequencing by channel composition and tensor by the tensor product of channels. The case denotes $[\![\mathscr{C}[h]]\!] \circ \Psi_i + \sum_{j\neq i} [\![h_j]\!] \circ \Psi_j$. Thus exactly one syntactic occurrence depends on the hole, and each branch continuation has the channel type stated by the rules. These contexts can use a hole on one instrument outcome and omit it on the others. They do not contain iso-level operations on that hole.

Proof of Corollary 6.2. Discarding the exact count removes every term with two different labels and leaves $\sum_k x_k \rho x_k^\dagger = \mathscr{M}(\rho)$. With a finite counter, exits whose counts agree modulo K carry the same label. Under the agreement condition, each contributes $b_k L u$. Factoring out the common data vector gives $R_K u = \xi_K \otimes Lu$, where $\xi_K = \sum_k b_k |k \bmod K\rangle$. For any unit input u, both $R_K u$ and Lu have norm one, so ξ_K does too. Discarding it gives $\mathscr{E}_L$. If $X = 0$, then $X' = 0$ and there is only one channel to compare.

Equal channels still agree when the caller keeps a finite reference: elementary tensor operators span the joint matrix algebra. They also agree after composition with a fixed channel or discard. In a case expression we can substitute equal channels on the unnormalized state of each outcome, then add the outputs if the caller forgets the outcome. Induction on the context now gives the same observations for both loops.

For the converse we exhibit tests available to the caller. On either space we can prepare basis states and, for each pair r, s, the states $(3|r\rangle + 4|s\rangle)/5$ and $(3|r\rangle + 4i|s\rangle)/5$. Their projectors, together with the basis projectors, span the matrix algebra: the two superpositions supply the real and imaginary off-diagonal parts. Thus if the channels differ, one of these inputs gives different outputs.

A basis measurement detects a difference on the diagonal. If the diagonals agree, projecting onto one of the two superpositions detects a nonzero real or imaginary part. Each preparation and test needs only a basis state or measurement and a two-level unitary over $\mathbb{Q}(i)$. □

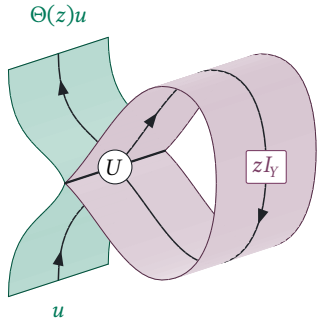
For an immediate identity bypass on a nonzero space Z, the first exit is $I_Z \oplus A$ and each later exit is $0 \oplus x_k$. A multiple of the first can vanish on Z only if it is zero. If all delayed exits vanish, $A^\dagger A = I$ by exit completeness. Since $A^\dagger A + C^\dagger C = I$, we get $C = 0$. Conversely, $C = 0$ makes every delayed exit zero. An isolated internal return port can still be present.

If the caller carries the count along while acting only on the data, it may discard the count at the end. Write H for the count register and τ for its joint state with the data. For each Kraus operator T of a branch on the data, $\mathrm{tr}_H((I_H \otimes T)\tau(I_H \otimes T)^\dagger) = T(\mathrm{tr}_H\,\tau)T^\dagger$. Summing over those operators gives the same unnormalized output whether the discard comes before or after the branch.

Why one finite clock does not suffice. The output interface matters. If a caller reads the count, even a fixed count independent of the data is an observable output. Conversely, a finite clock may hide a failure of agreement. For $m > 0$, use the permutation that fixes `exit 0` and cycles through $\texttt{exit}\,1 \mapsto \texttt{back}\,q_1 \mapsto \cdots \mapsto \texttt{back}\,q_m \mapsto \texttt{exit}\,1$, with m distinct return values $q_1, \ldots, q_m$. Its only nonzero exits are $x_0 = |0\rangle\langle 0|$ and $x_m = |1\rangle\langle 1|$. The m-clock assigns label 0 to both exits. Ignoring it preserves $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$, whereas a measurement on every return dephases $|+\rangle$ in that basis. Checking just this clock would accept the replacement incorrectly.

E The derivative and its average

In `guardRound`, write $s = 4/5$ and $t = 3/5$. The phase gate receives s on the first return, stz on the next, and st^2z^2 on the third. Summing over all returns gives $F_z = s/(1 - tz)$. For a general body the corresponding series is $F_z = C + zD_1C + z^2D_1^2C + \cdots$. Here D_1 and B_1 are the restrictions of D and B to $Y_1 = Y_0^\perp$, where $Y_0 = \bigcap_{j\ge 0} \ker BD^j$ consists of the states that never emit. By Theorem 3.1, C takes values in Y_1 and powers of D_1 decay geometrically. The series therefore gives $F_z = (I - zD_1)^{-1}C$ even at a phase where $I - zD$ on all of Y is singular.



	$j=0$	$j=1$	$j=2$
$k=0$	T_{00}	$z^{-1}T_{01}$	$z^{-2}T_{02}$
$k=1$	zT_{10}	T_{11}	$z^{-1}T_{12}$
$k=2$	z^2T_{20}	zT_{21}	T_{22}

$$T_{kj} = x_k \rho x_j^\dagger$$

$$\Theta_{\mathrm{rot}}(z) = \frac{t - z}{1 - tz},$$

$$\Theta_{\mathrm{bypass}}(z) = \mathrm{diag}(1, \Theta_{\mathrm{rot}}(z)).$$

$$\Theta(z) = \sum_{k\ge 0} z^k x_k = A + zB_1(I - zD_1)^{-1}C, \qquad L = \Theta(1).$$

Fig. 18. Each return adds a factor z at left. In the channel at right, exit pair k, j carries $z^{k-j}T_{kj}$. Averaging over the circle retains only the orange diagonal. The same cancellation applies to later exits. The displayed families Θ_{rot} and Θ_{bypass} belong to `guardRound` and `bypassRound`.

Proof of Theorem 7.1. We use the same body at two return phases. On Y_1 the wire equation is $F_z = C + zD_1F_z$, so a call takes (u, zF_zu) to $(\Theta(z)u, F_zu)$. Comparing this call with one on (v, wF_wv) at phase w, we can equate the input and output inner products because U_1 is unitary. With the inner product linear in its second argument, this gives

$$\langle v, u\rangle + \bar{w}z\langle F_wv, F_zu\rangle = \langle \Theta(w)v, \Theta(z)u\rangle + \langle F_wv, F_zu\rangle.$$

Collecting the two occurrences of the returning inner product gives $I - \Theta(w)^\dagger\Theta(z) = (1-\bar{w}z)F_w^\dagger F_z$. On the open unit disk this is the transfer identity for the characteristic function of the unitary colligation [10] [6, Eq. (4.10)].

The decay in Theorem 3.1 makes F_z and $\Theta(z)$ analytic on a disk extending beyond the unit circle. Fix $|w| = 1$ and differentiate the identity in z. At $z = w$ the factor $1 - \bar{w}z$ vanishes, taking the term with F_z' with it. This leaves $\Theta(w)^\dagger\Theta'(w) = \bar{w}F_w^\dagger F_w$. Since $dw/d\varphi = iw$, this gives $Q(w) = F_w^\dagger F_w$.

To average this weight over phase, expand $F_z = \sum_{j\ge 0} z^j D_1^j C$. In $F_z^\dagger F_z$ the term indexed by m, n has phase z^{n-m}. Integration around the circle keeps only terms with $m = n$, just as it does for the channel in Figure 18. The geometric bound permits termwise integration, so the average is $\sum_j T_j$, where $T_j = (D_1^jC)^\dagger D_1^jC$ is the weight still returning after call $j+1$.

An exit after k returns contributes to the returning weight on k calls. Weighting $x_{j+1}^\dagger x_{j+1} = T_j - T_{j+1}$ by $j+1$ and summing through $j = p-1$ gives $\sum_{k\le p} kx_k^\dagger x_k = \sum_{j<p} T_j - pT_p$. Geometric decay sends pT_p to zero, so the phase average $\sum_j T_j$ equals N. On a unit input, its expectation gives the mean number of returns.

Taking traces gives $\operatorname{tr} N = \sum_j \operatorname{tr} T_j$. Cyclicity puts each trace on Y_1, where the return row gives $CC^\dagger = I - D_1D_1^\dagger$. Substituting this difference cancels consecutive terms:

$$\sum_{j=0}^{p-1} D_1^jCC^\dagger D_1^{\dagger j} = I_{Y_1} - D_1^pD_1^{\dagger p} \longrightarrow I_{Y_1}.$$

Taking the trace gives $\operatorname{tr} N = \dim Y_1$. When $Y_1 = 0$, $F_z = 0$, so all three quantities vanish. □

For monitored first returns, identify the external output unitarily with the input X. An exit after k returns takes $k+1$ calls, so counting calls multiplies $\Theta(z)$ by one more z. The relative phase derivative of $z\Theta(z)$ is therefore $I + Q(z)$. For a single monitored state, Grünbaum et al. express its mean first return time as the winding of the first-arrival amplitude [23, § 4, proof of Thm. 2]. The mean number of calls is $1 + \langle u, Nu\rangle$ [9, § 3, Eq. (8)]. A vector orthogonal to the smallest reducing subspace containing X stays orthogonal to X under every power of U. It must therefore be internal and never emit. Conversely, $0 \oplus Y_0$ reduces U by Theorem 3.1 and is orthogonal to X. This identifies the smallest reducing subspace as $X \oplus Y_1$. The mean on u is the quadratic form of $I + N$, so its sum over an orthonormal basis is $\dim X$ times its uniform unit-sphere average. The monitored return times therefore sum to $\dim X + \dim Y_1$ [9, Thm. 3.2, Eq. (9)]. Removing one initial call per basis vector gives $\operatorname{tr} N = \dim Y_1$.

Phase invariance and agreement. On a nonempty external type, the phase family also characterizes agreement. If $x_k = b_kL$ for every exit, then $\Theta(z) = f(z)L$ with $f(z) = \sum_k b_kz^k$. For $|z| = 1$ the marked body is unitary, so its trace $\Theta(z)$ preserves the norm of any unit input. Since L does too, $|f(z)| = 1$. This scalar phase cancels in the data channel. Conversely, if those channels are all equal, they equal $\mathscr{E}_L$ because $\Theta(1) = L$. Their average is $\mathscr{M}$ by (7.1), so $\mathscr{M} = \mathscr{E}_L$. Thus agreement says exactly when changing the return phase leaves the data channel unchanged.

F Clause programs and their observations

Two scalar loops in sequence. To write the two copies of `guardRound` from Section 8 as one body, we give each return wire its own basis value. The open matrix acts on $\mathbb{C} \oplus (\mathbb{C} \oplus \mathbb{C})$. Both wires receive the same phase z, so the response in sequence is $f(z)^2$, where $f(z) = (t-z)/(1-tz)$, $t = 3/5$ and $s = 4/5$. To read its columns, let u enter the first rotation and y_1, y_2 enter the two return ports. The first rotation sends $tu - sy_1$ into the second. The external output is therefore $t(tu - sy_1) - sy_2$, and the two return outputs are $su + ty_1$ and $s(tu - sy_1) + ty_2$. Reading off the coefficients of u, y_1, y_2 in turn gives the three clauses.

```
round twoSections : Unit + Qubit = {
  exit unit ↔ [9/25 . exit unit + 4/5 . back 0 + 12/25 . back 1];
  back 0 ↔ [-12/25 . exit unit + 3/5 . back 0 + -16/25 . back 1];
  back 1 ↔ [-4/5 . exit unit + 3/5 . back 1]
}

trace twoSections on unit
```

The two loops return $|*\rangle$, since $f(1)^2 = 1$. A period-two counter returns label 0 with certainty. At phase 1 the even and odd exits add, and at phase -1 the odd exits change sign. Both sums are one, so the odd exits sum to zero and the even exits sum to one.

Shared and separate counts. Measuring each loop can reveal a distinction that the total alone misses. For example, let the first body send $|0\rangle$ straight through and delay $|1\rangle$ by one return. Let the second do the reverse, with unit amplitude on each path. On input $|+\rangle$, separate counters leave the joint state $(|0,1\rangle \otimes |0\rangle + |1,0\rangle \otimes |1\rangle)/\sqrt{2}$. The two records are orthogonal, so tracing them out leaves $I/2$. A shared counter carries $|1\rangle$ on both paths, giving $|1\rangle \otimes |+\rangle$. In the combined body the only nonzero exit map is therefore $x_1 = I$, and measuring its outer tag also preserves $|+\rangle$.

F.1 A flip on every call

In `flipRound`, each call rotates the guard by R_θ and flips the data qubit. Both clauses therefore call `X` before forming their two outputs.

```
X :: Qubit ↔ Qubit
X = { 0 ↔ 1; 1 ↔ 0 }

round flipRound : Qubit + Qubit = {
  exit x ↔ let y = X x in [ 3/5 . exit y + 4/5 . back y];
  back x ↔ let y = X x in [-4/5 . exit y + 3/5 . back y]
}
```

On `exit 0`, the first clause binds $x = 0$ and the call to `X` gives $y = 1$. Of its two outputs, only $(4/5)|11\rangle$ reaches the second call. This time the `back` clause flips the data to 0, so the part that leaves has amplitude $(4/5)(-4/5) = -16/25$. The returning part is $(12/25)|10\rangle$, so a third call emits $-(48/125)|01\rangle$ and sends $(36/125)|11\rangle$ back. With the guard first, the open matrix is $R_\theta \otimes \mathrm{X}$.

The exit maps of `flipRound`. Put $t = 3/5$ and $s = 4/5$. The two clauses of `flipRound` give $A = t\mathrm{X}$, $C = s\mathrm{X}$ on entry and $B = -s\mathrm{X}$, $D = t\mathrm{X}$ on return. Hence $x_0 = t\mathrm{X}$ and $x_1 = BC = -s^2 I$. The data flips twice before that second exit. In general $x_{k+1} = -s^2 t^k \mathrm{X}^{k+2}$. Grouping equal data gates gives two

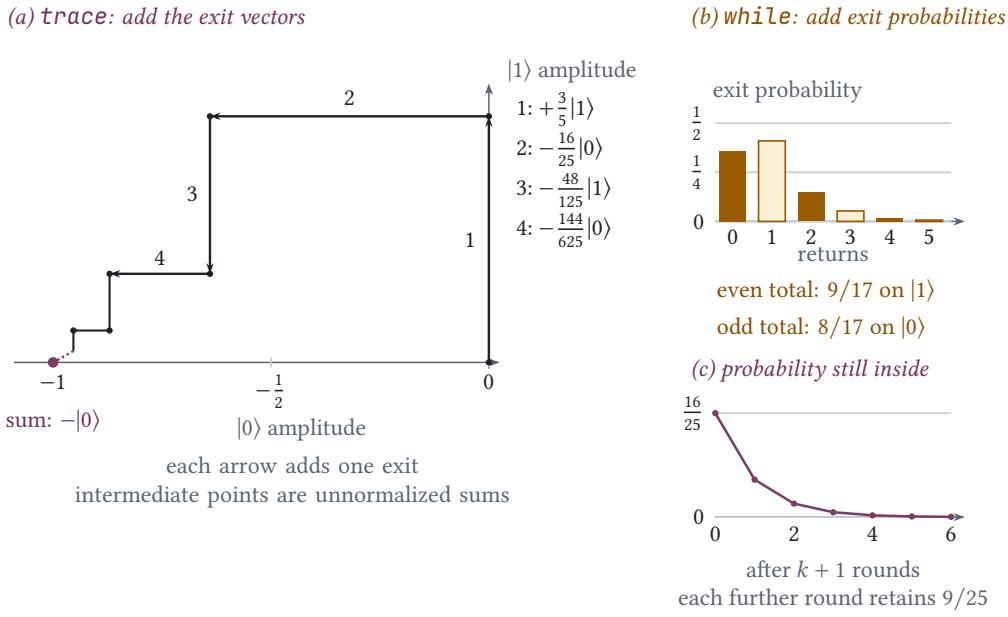


Fig. 19. On input $|0\rangle$, the left panel adds one exit amplitude per call, starting with call one. The bars at right give the exit probabilities by number of returns, starting at zero for the first call.

ordinary geometric series,

$$\underbrace{t - s^2 t \sum_{j\geq 0} t^{2j} = 0,}_{\text{coefficient of X}} \qquad \underbrace{-\, s^2 \sum_{j\geq 0} t^{2j} = -1,}_{\text{coefficient of } I} \qquad s^2 = 1 - t^2.$$

Thus the whole coherent loop returns $-u$. In particular it can never return data 1 on input 0. With an exact count the exits that return 0 on this input have odd counts, with amplitudes $-s^2, -s^2t^2, \ldots$. Their squared magnitudes sum to $s^4/(1-t^4) = 8/17$. The complementary probability $9/17$ goes to 1. These are the two observations in Figure 19. If we keep only parity, the even counts carry the same label and their amplitudes cancel by the first geometric sum. The odd counts carry the other label and add to -1. Under measurement, exits at different counts contribute separately to the probabilities above.

In Figure 5, eliminating `back 0` leaves a self-loop of amplitude $9/25$ on `back 1`. Input 0 sends $4/5$ there, so successive returns sum to $(4/5)/(1-9/25) = 5/4$. It contributes $(-4/5)(5/4) = -1$ to output 0 and $(-12/25)(5/4) = -3/5$ to output 1, canceling the immediate $3/5$. Interchanging the bits gives the second column of $-I$.

Kickback with a returned ancilla. In `flipRound` the payload changes on every call. We now give the payload two qubits and arrange for the closed loop to flip the data while returning the ancilla. On each call we rotate the guard by $\pi/4$ and apply the same gate M to the payload. Write $c = 1/\sqrt{2}$. The immediate exit is cM, the first delayed exit is $-c^2M^2$, and each further return adds a factor cM. Since $2c^2 = 1$, their sum is $f(M) = cM - c^2M^2(I - cM)^{-1} = M(cI - M)(I - cM)^{-1}$.

Write $J = \mathsf{X}$ and let H be Hadamard. With $G = HJ$ and $\Pi_\pm = (I \pm J)/2$, we choose

$$M = G \otimes \Pi_+ + JG \otimes \Pi_-.$$

In ISO this is an operation controlled in the J basis of the data. We change that basis with H, apply G or JG to the ancilla according to the resulting data bit, then undo the basis change.

To see which qubit receives the sign, look at $f(M)$ in each sector. For $M = G$, the identity $G^2 = 2cG - I$ gives $G(cI - G) = I - cG$. The numerator cancels the denominator, so $f(G) = I$. For $M = JG$, we have $(JG)^2 = I$, so the numerator is now $cJG - I = -(I - cJG)$ and $f(JG) = -I$. Both denominators are invertible since $c < 1$ and both gates are unitary. Since the ancilla comes back in either sector, the relative sign acts on the data, giving $I \otimes (\Pi_+ - \Pi_-) = I \otimes J$ on every joint input.

In the basis of the two payload bits, the projector sum gives

$$M|0,d\rangle = \frac{|0,1-d\rangle - |1,1-d\rangle}{\sqrt{2}}, \qquad M|1,d\rangle = \frac{|0,d\rangle + |1,d\rangle}{\sqrt{2}}.$$

The data bit flips in the first case and stays in the second. Expanding the guard rotation now gives the clauses below. Each nonzero coefficient of M is $\pm 1/\sqrt{2}$, and the guard contributes another $1/\sqrt{2}$, leaving only coefficients $\pm 1/2$. Thus no rounded Hadamard coefficient is needed in this source.

```
round kickbackExact : (Qubit x Qubit) + (Qubit x Qubit) = {
  exit ⟨0, 0⟩ ↔ [1/2 . exit ⟨0, 1⟩ + -1/2 . exit ⟨1, 1⟩
                + 1/2 . back ⟨0, 1⟩ + -1/2 . back ⟨1, 1⟩];
  exit ⟨0, 1⟩ ↔ [1/2 . exit ⟨0, 0⟩ + -1/2 . exit ⟨1, 0⟩
                + 1/2 . back ⟨0, 0⟩ + -1/2 . back ⟨1, 0⟩];
  exit ⟨1, 0⟩ ↔ [1/2 . exit ⟨0, 0⟩ + 1/2 . exit ⟨1, 0⟩
                + 1/2 . back ⟨0, 0⟩ + 1/2 . back ⟨1, 0⟩];
  exit ⟨1, 1⟩ ↔ [1/2 . exit ⟨0, 1⟩ + 1/2 . exit ⟨1, 1⟩
                + 1/2 . back ⟨0, 1⟩ + 1/2 . back ⟨1, 1⟩];
  back ⟨0, 0⟩ ↔ [-1/2 . exit ⟨0, 1⟩ + 1/2 . exit ⟨1, 1⟩
                + 1/2 . back ⟨0, 1⟩ + -1/2 . back ⟨1, 1⟩];
  back ⟨0, 1⟩ ↔ [-1/2 . exit ⟨0, 0⟩ + 1/2 . exit ⟨1, 0⟩
                + 1/2 . back ⟨0, 0⟩ + -1/2 . back ⟨1, 0⟩];
  back ⟨1, 0⟩ ↔ [-1/2 . exit ⟨0, 0⟩ + -1/2 . exit ⟨1, 0⟩
                + 1/2 . back ⟨0, 0⟩ + 1/2 . back ⟨1, 0⟩];
  back ⟨1, 1⟩ ↔ [-1/2 . exit ⟨0, 1⟩ + -1/2 . exit ⟨1, 1⟩
                + 1/2 . back ⟨0, 1⟩ + 1/2 . back ⟨1, 1⟩]
}

trace kickbackExact on 3/5 . ⟨0, 0⟩ + 4/5 . ⟨1, 1⟩
```

The input is entangled between ancilla and data. Its output is $3|0,1\rangle/5 + 4|1,0\rangle/5$, exactly the result of applying J to the second component. The same identity on arbitrary joint inputs follows from the two-sector calculation above.

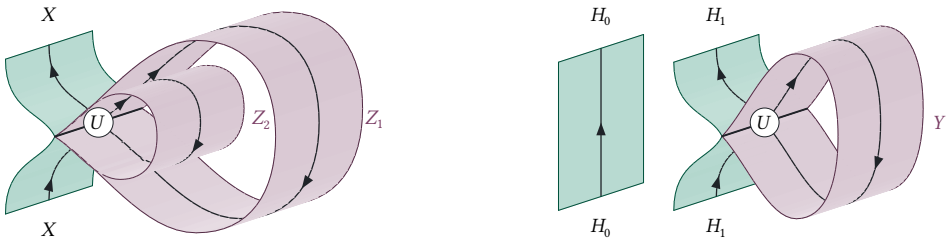
G Reading the feedback diagrams

Reading a sheet diagram from bottom to top, we follow each summand on its own surface. Along the depth of that surface run the wires of its tensor factors. This is why registers on one sheet can be used together, while different sheets carry alternative components. Where the surfaces meet, the dark seam labeled U acts on both summands, even where the returning sheet hides part of the external wire. Bending an entire surface back to its input takes the direct-sum trace over that summand. The sheet calculus supplies sum, tensor and composition [15, Defs. 16, 18–20, 22].

The two guard values label the summands of $X \oplus Y$. The trace closes the return summand Y. Hiding a register in Section 5 takes the partial trace over its tensor factor.

The finite trace satisfies the trace laws even for a body with a trapped return [3, Cor. 3.3.16] [5, Thms. 7–8]. The measured loop in Appendix C has a least fixed point. For the coherent loop, deriving a fixed-point operator from the trace by the cartesian construction would require a diagonal [24, Thm. 3.1]. For direct sum, $u \mapsto (u, u)$ doubles the squared norm of every nonzero u, so it cannot be a unitary in our language.

The two sheets below show how the trace acts on a nested loop and on a loop beside a bypass. Both equations use canonical reassociations of the summands.



$\mathrm{Tr}^{Z_1}(\mathrm{Tr}^{Z_2} U) = \mathrm{Tr}^{Z_1 \oplus Z_2} U$ $\qquad$ $I \oplus \mathrm{Tr}^{Y} U = \mathrm{Tr}^{Y}(I \oplus U)$

Fig. 20. At left, both return ports of U are closed. At right, only the input on H_1 enters the loop.

The right sheet gives the block form used for `bypassRound`. The bypass contributes only to the immediate exit, whose block is $I \oplus x_0$. Each later exit is $0 \oplus x_k$, so their sum is $I \oplus L$. On the left, closing the two ports in either order solves both wire equations, as in Proposition 3.2.

G.1 The backward wire in the Int construction

In the loop diagram a wire runs backwards from the returning output to the next input. To make this drawing into a typed composition, the Int construction pairs each forward type P^+ with a backward type P^-. A map from P to Q can then consume P^+ from its left and Q^- from its right, so its underlying unitary has type $P^+ \oplus Q^- \to Q^+ \oplus P^-$. Figure 21 draws these ports.

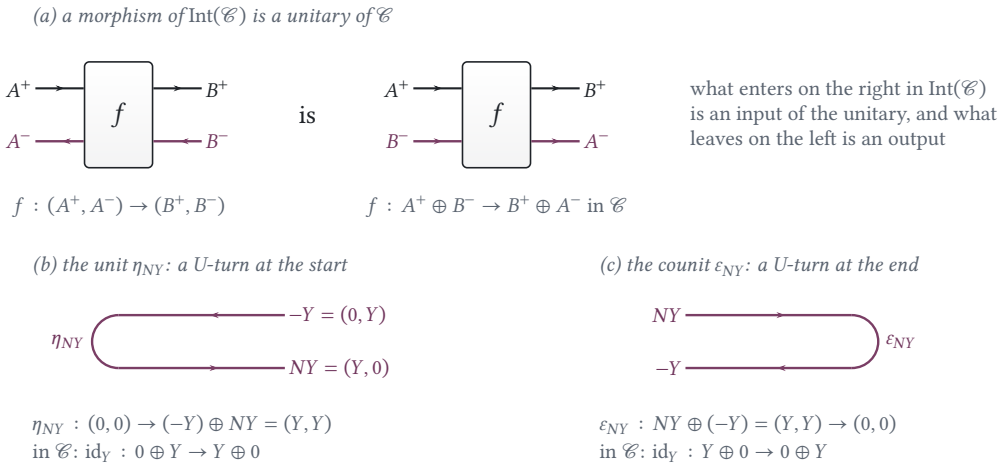


Fig. 21. Objects in the compact completion have forward and backward components. Cups and caps bend a port within that completion.

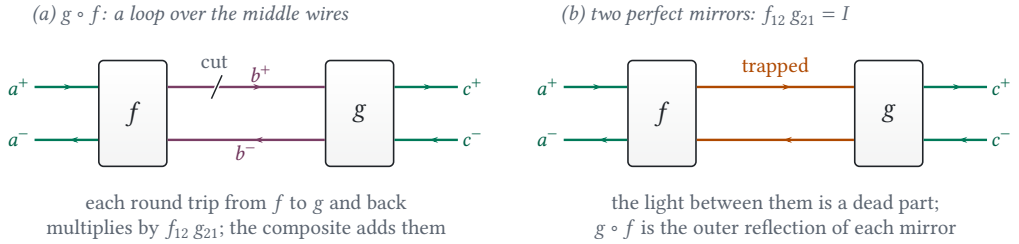


Fig. 22. Composition closes the middle ports. Between two perfect mirrors, a round trip returns the light unchanged and never reaches an outer port.

In Figure 22, f sends b^+ to g and g sends b^- back to f. On external inputs a^+ and c^- the two middle amplitudes satisfy

$$b^+ = f_{11}a^+ + f_{12}b^-, \qquad b^- = g_{21}b^+ + g_{22}c^-.$$

Follow b^+ once around the middle wires. The part reflected by g returns to f as $g_{21}b^+$, and f sends $f_{12}g_{21}b^+$ past the cut again. Substitution in the first equation gives

$$b^+ = \left(f_{11}a^+ + f_{12}g_{22}c^-\right) + f_{12}g_{21}b^+.$$

The parentheses collect what the two external inputs send to the cut. This is the wire equation of Section 3, with $y = b^+$ and $D = f_{12}g_{21}$. Once we solve it, f and g give the external outputs a^- and c^+. When $I - D$ is invertible this elimination gives the scattering matrix of the pair, the Redheffer star product [21, Figs. 10–11].

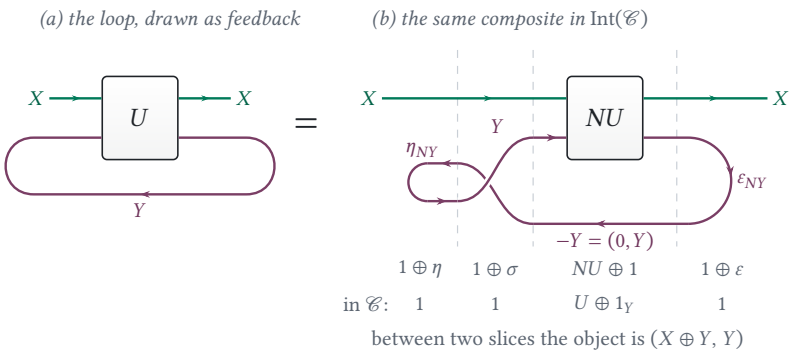


Fig. 23. Both diagrams close Y in $U : X \oplus Y \to X \oplus Y$. The cap takes the return onto the backward wire $(0, Y)$ and the cup brings it to the input.

The fully faithful embedding $N(X) = (X, 0)$ gives X no backward port. Bending Y around NU identifies its returning output $Cu + Dy$ with its returning input y. Thus the compact closed trace in Figure 23 gives $N(L)$ by Theorem 3.1 [31] [35, Thms. 5.15–5.16].

The sum in Int reverses the order of negative components:

$$(P^+, P^-) \oplus (Q^+, Q^-) = (P^+ \oplus Q^+, Q^- \oplus P^-).$$

Reversing the ports of $(Y, 0)$ gives its dual $(0, Y)$. The unit and counit are identity wires on Y with their ports rearranged. Composing two maps plugs the forward and backward middle ports together

and takes the original trace. With $\eta_{NY} : 0 \to (-Y) \oplus NY$ and $\varepsilon_{NY} : NY \oplus (-Y) \to 0$, the four maps compose as follows

$$N(\mathrm{Tr}^{Y} U) = (I_{NX'} \oplus \varepsilon_{NY})(NU \oplus I_{-Y})(I_{NX} \oplus \sigma_{-Y,NY})(I_{NX} \oplus \eta_{NY}).$$

Figure 23(b) draws the case $X' = X$. Read its four regions from left to right. The cup opens the two ports and the swap puts them in the order $NU \oplus I_{-Y}$ expects. In each direction Y moves past the zero space, so the underlying swap is $I_{Y \oplus Y}$, or $I_{X \oplus Y \oplus Y}$ with the external wire. The cap then joins the ports. Both bends belong to the compact completion.

H A ring realization of the same exits

In Figure 7, light stays in its waveguide with amplitude $t = 3/5$ and crosses to the other waveguide with amplitude $is = 4i/5$ [26, Eq. (1)]. The light field extends a little beyond each guide, so the fields overlap across the narrow gap [2]. Light crossing into the ring from u keeps its direction of travel. It leaves the gap as d, comes around again as y, and the part that crosses back joins the direct light on its way to v.

We can see where the i comes from by checking the columns, just as we did for `guardRound`. With through amplitude $t = 3/5$ and the same crossing amplitude x in both directions, the columns are (t, x) and (x, t). Their inner product $t(x + \bar{x})$ must vanish, so x is purely imaginary. Here we choose $x = is$. This coupler is the rotation from `guardRound` in a different basis on the return wire, $(1 \oplus i)R_{\theta}(1 \oplus (-i))$. To leave after one trip, light crosses into the ring with amplitude is and crosses back with amplitude is. Each further trip adds a factor t for staying in the ring at the coupler. The exit after $k + 1$ trips thus has coefficient $(is)t^{k}(is) = -s^{2}t^{k}$. Hence the coupler and the rotation have the same exit on every round and the same response $\Theta_{\mathrm{ring}}(z) = (t - z)/(1 - tz)$ at round-trip phase z. In the two clauses below, `(0 :+ 1)` is the coefficient i.

```
round ringCoupler : Unit + Unit = {
  exit unit ↔ [3/5 . exit unit + (0 :+ 1) . 4/5 . back unit];
  back unit ↔ [(0 :+ 1) . 4/5 . exit unit + 3/5 . back unit]
}

trace ringCoupler on unit
```

With no added phase the ring returns $-|*\rangle$.

To read both panels of Figure 9 from this program, we also need to say when each exit reaches the straight guide. Write $u(T)$ for the incident field, with time measured relative to the direct path. The copy that makes k trips arrives $k\tau_0$ later with the amplitude of exit k. The field at the marked port is therefore

$$v(T) = t\,u(T) - s^{2} \sum_{k \geq 1} t^{k-1} u(T - k\tau_0).$$

Raymer and McKinstrie give this train of delayed copies with the opposite overall output sign [40, Eqs. (30)–(33)]. If the pulse ends before the first return, the delayed copies arrive separately. For a steady wave that completes a whole number of cycles in time τ_0, shifting it by any multiple of τ_0 gives back $u(T)$. Their coefficients sum to $-s^{2}/(1 - t) = -8/5$. Adding the direct $3/5$ gives the sign returned by the program [2, Eqs. (6)–(7)].

To realize `bypassRound`, match the bypass to the straight-through path in phase and travel time. Prepare the external qubit in $|+\rangle$, run the two paths, and apply H to compare them. At this frequency the delay around the ring has phase factor $z = 1$, so the ring contributes -1 against the bypass amplitude 1 and the difference port has probability $|1 - (-1)|^{2}/4 = 1$.

For a short pulse, only the direct field arrives with the bypass. Their amplitudes at the difference port are $-t/2$ and $1/2$, so the first bin contributes $(1-t)^2/4$. Each later bin contains just one delayed exit. The two splitters give its amplitude a factor $1/2$, and we add the resulting probabilities.

$$\Pr(b=1) = \frac{1}{4}\left((1-t)^2 + \sum_{k\geq 1} s^4 t^{2(k-1)}\right) = \frac{1}{4}\big((1-t)^2 + 1 - t^2\big) = \frac{1}{5}.$$

These are the two probabilities for `bypassRound` in Table 1. The ring alone returns scalar data in either experiment. Including the bypass in the body lets the immediate exit carry both paths, revealing their relative sign.

A pulse contains a range of frequencies, each with its own round-trip phase $\omega\tau_0$. Suppose the incident photon has the same path state $|+\rangle$ at every frequency, with frequency distribution $p(\omega)$. A detector that counts every arrival at the difference port without distinguishing frequencies therefore averages the single-frequency probabilities, giving $\int p(\omega)|1-\Theta_{\mathrm{ring}}(e^{i\omega\tau_0})|^2\, d\omega/4$. The component at resonance goes entirely to the difference port. For a component whose round-trip phase is δ away from a whole turn, the probability of reaching the other port, for fixed $|t|<1$, is

$$\frac{|1+\Theta_{\mathrm{ring}}(e^{i\delta})|^2}{4} = \frac{(1+t)^2\sin^2(\delta/2)}{(1-t)^2+4t\sin^2(\delta/2)} = \frac{Q(1)^2}{4}\delta^2 + O_t(\delta^4).$$

If the spectrum lies in $|\delta| \leq h$ and $\sigma^2 = \mathbb{E}[\delta^2]$, then $\mathbb{E}[\delta^4] \leq h^2\sigma^2$. As $h \to 0$ the mean probability of reaching the other port is therefore $Q(1)^2\sigma^2/4 + o(\sigma^2)$. If the round-trip phase is uniform, averaging the full response over the spectrum gives the measured loop by (7.1).

Section 4 gives the switch-on field $-1+(1+t)t^n$ after n trips. Near $t=1$ it takes $-1/\log t \sim 1/(1-t)$ trips to reduce the difference from the limiting amplitude -1 by a factor of e. For the steady output, we can also ask how its phase changes with frequency. The circulating intensity for unit input is $Q(1)$ [26, Eq. (7)]. Since $\varphi = \omega\tau_0$, we have $d\varphi/d\omega = \tau_0$. With a frequency-independent coupler this gives $-i\Theta^\dagger d\Theta/d\omega = \tau_0 Q(1)$ [44, 50]. For a wave packet, the corresponding dwell time involves averaging the Wigner–Smith matrix over the packet's energy spectrum [18, Eqs. (34)–(35)].

I Observing the response

We put a phase on each return by composing β with $I_b \oplus P_z^c$, where $P_z^c = \{s \leftrightarrow z\cdot s\}_{s\in\mathscr{B}(c)} : c \leftrightarrow c$ and $|z|=1$. With the bypass this gives the program

$$H; \mathsf{trace}_1\big(\texttt{bypassRound}; (I_{\texttt{Qubit}} \oplus P_z^1)\big); H.$$

The bypass exits at once and the rotation makes one return on average. Since only the rotation returns, $N_{\mathrm{bypass}} = \mathrm{diag}(0,1)$, giving mean $1/2$ on $|+\rangle$. The final Hadamard at phase $z = e^{i\varphi}$ gives the probability shown below.

$$\Pr(b=1) = |1-\Theta_{\mathrm{rot}}(z)|^2/4$$

φ	0	$\pi/2$	π
$\Theta_{\mathrm{rot}}(e^{i\varphi})$	-1	$(15-8i)/17$	1
$\Pr(b=1)$	1	$1/17$	0

Averaging this test over the phase circle gives the measured probability $1/5$ by (7.1).

Comparing two nearby phases. The bypass compares the loop with a straight path. To distinguish the circulating weights 4 and $1/4$ of `guardRound` (return amplitudes 2 at $z=1$ and $1/2$ at $z=-1$), we compare the loop at two nearby phases. A qubit $H_r = \mathbb{C}^2$ accompanies the data on both wires. By distributivity, $H_r \otimes (X \oplus Y) \cong (H_r \otimes X) \oplus (H_r \otimes Y)$. With $P_\delta = \mathrm{diag}(1, e^{i\delta})$ the probe controls

the following body at angle $\delta \in \mathbb{R}$

$$W_{z,\delta} = \big(I_{H_r \otimes X'} \oplus (P_\delta \otimes z I_Y)\big)(I_{H_r} \otimes U).$$

Because the body preserves the reference bit, the kth exit is $z^k x_k$ on branch zero and $(ze^{i\delta})^k x_k$ on branch one. Summing the exits gives $|0\rangle\langle 0|_r \otimes \Theta(z) + |1\rangle\langle 1|_r \otimes \Theta(ze^{i\delta})$.

On input $|+\rangle \otimes u$ with $\|u\| = 1$, this gate returns

$$\frac{|0\rangle_r \otimes \Theta(z)u + |1\rangle_r \otimes \Theta(ze^{i\delta})u}{\sqrt{2}}.$$

The caller applies $S^\dagger$ to put a factor $-i$ on the second branch, then H to bring the branches together. At outcome 0 the data vector is $(\Theta(z)u - i\Theta(ze^{i\delta})u)/2$. At outcome 1 the minus becomes a plus. Subtracting the two probabilities cancels the squared norm of each branch and leaves $\mathrm{Im}\langle \Theta(z)u, \Theta(ze^{i\delta})u\rangle$. By the definition of $Q(z)$, this difference is $\delta\langle u, Q(z)u\rangle + O(\delta^2)$.

Scoring zero as $+1$ and one as -1 gives variance $1 - \mu^2$ for mean score μ. For n independent runs, the average score divided by δ estimates the weight with variance $(1 - \mu^2)/(n\delta^2)$ and bias $O(\delta)$. At a fixed body and phase, choosing δ proportional to ϵ and n proportional to ϵ^{-4} makes both bias and standard error $O(\epsilon)$. The constants depend on the local variation of the response.

For phases 1 and i, `guardRound` returns -1 and $(15 - 8i)/17$. Their overlap $(-15 + 8i)/17$ gives a probability difference of $8/17$, so the caller reads zero with probability $25/34$ and one with $9/34$. Dividing by the quarter turn $\pi/2$ gives $16/(17\pi)$. As we bring the second phase back toward 1, this quotient tends to the local weight 4.